Parental Leave and Gender Gaps in the Labor Market*

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Abstract

Parental leave mandates protect workers' jobs but impose costs on the firms that must grant leave. We study Taiwan's 2009 parental leave reform. Before the reform, only firms with 30 or more employees had to grant job-protected leave. The reform extended job protection to all firms and introduced paid leave financed by employment insurance for workers at firms of every size. Using population-wide employer–employee records and a difference-in-differences design, we compare newly covered small firms with always-covered large firms. Relative to large firms, small firms reduced the share of women of childbearing age among their new hires, while the share of men of the same age fell much less. These women became less likely to move between small firms, with no clear change in their wage growth relative to men, and mothers at small firms became less likely to be employed in the first years after birth, while fathers' employment did not fall. In a Burdett–Mortensen search model in which gender equal-pay rules prevent firms from paying women of childbearing age less than other workers, a cost of covering an absence that falls with the number of coworkers explains the response to the reform, and the required cost may be as small as the direct cost of an absence. A much larger cost would be needed to explain why these women find jobs more slowly than other workers at firms of all sizes, suggesting that this broader gap also reflects labor supply and other costs that employers attach to hiring them.

JEL Codes: J13, J16, J18, J23

Keywords: Parental leave; Child penalty; Gender gap; Employer responses

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1 Introduction

Although the gender gap in wages has narrowed considerably, large gender disparities in labor market outcomes persist in many countries, and a growing body of research identifies the arrival of children as a main source of these disparities (Adda et al., 2017; Kleven et al., 2024; Kuziemko et al., 2018). After childbirth, women are far more likely than men to cut their hours, move into part-time work, or leave the labor force altogether, while men's employment is largely unaffected. Parental leave is one of the main tools governments use to soften this penalty, yet the evidence on its effects is mixed, and some reforms appear to have worsened mothers' labor market outcomes (Dahl et al., 2016; Kleven et al., 2024).

Most of this literature studies leave from the labor supply side, asking how an entitlement changes mothers' decisions to take time off and return to work (Cortés and Pan, 2023; Kleven et al., 2025). However, a leave mandate also imposes costs on employers. A firm that must grant leave has to hold the position open, cover the work of the absent employee, and bear the frictions of finding a replacement, which gives it an incentive to change whom it hires. Understanding why leave policies may fail to narrow gender gaps requires knowing how firms respond and how their response feeds back into workers' outcomes. Yet the two sides of the market are usually studied separately, and few studies examine both with the same policy variation (Di Nola et al., 2026).

This paper studies both sides of the market using a reform that changed which firms are required to grant leave. Before 2009, Taiwan offered unpaid, job-protected parental leave only to employees of firms with 30 or more employees. The 2009 reform extended job protection to employees of all firms and introduced paid leave financed by employment insurance for eligible workers at firms of every size. Using linked employer–employee tax records for the population of Taiwan from 2006 to 2017, we compare small firms that had to grant job-protected leave for the first time with large firms that were covered both before and after the reform. We find that small firms shifted their hiring away from women of childbearing age, that these women became less likely to move between small firms, and that mothers at small firms became less likely to be employed in the first years after birth. We then use a search model to infer what these responses imply about the cost that firms attach to leave.

Because paid leave became available at firms of every size, our difference-in-differences design compares firms that gained both job protection and paid leave with firms that gained only paid leave. Under the assumption that paid leave would have affected small and large firms in the same way, the comparison identifies the effect of extending job protection to small firms when paid leave is available, including any interaction between the two. Before the reform, a firm near the threshold could have

avoided the obligation by staying below 30 employees. We find no bunching in the firm-size distribution below 30 (Figure 9), so a classification based on employment before the reform is unlikely to reflect firms sorting to avoid coverage. The absence of bunching suggests that firms kept growing past the threshold and responded to the cost of leave by changing whom they hired.

Other countries also tie leave protection to firm size. In the United States, the Family and Medical Leave Act of 1993 guarantees twelve weeks of unpaid, job-protected leave only to employees of firms with at least fifty workers. In Germany, the core entitlement (*Elternzeit*) is universal, but the right to work part-time during leave applies only in firms with more than fifteen employees. How firms respond when such a threshold changes is relevant well beyond Taiwan.

We compare small and large firms before and after the reform and exclude 2009, because the reform took effect in May of that year. For firms, we estimate event studies of the share of new hires and the share of employees who are women aged 20 to 39, with industry-by-year fixed effects. To separate a response to women from a general shift away from young workers, we also use the gap between the shares of women and men of the same age as the outcome, which gives a triple-difference estimate. For workers, we estimate how the reform changed the probability that women aged 20 to 39 at small firms move to another small firm or to a large firm, relative to women at large firms, and how it changed their wage growth relative to men of the same age. Finally, we compare the employment and earnings of mothers whose employer before birth was a small firm with those of mothers at large firms, for births before and after the reform, separately for each year since birth. The analysis yields three sets of results.

First, small firms shifted their hiring away from women of childbearing age. Relative to large firms, the share of new hires who are women of childbearing age (20–39) began to fall in 2010 and kept falling, reaching about 2 percentage points below its pre-reform level by 2016. The share of young men among new hires fell by much less, so most of the decline is specific to women. As these hiring decisions accumulated, the workforce of small firms shifted toward older women. The share of women aged 20–39 fell by 3.5 percentage points, while the share of women aged 40–59 rose. The decline occurs at small firms of every size and is largest at the smallest ones.

Second, women of childbearing age at small firms lost job mobility without a clear change in pay. They became less likely to move to another small firm, by 3.3 percentage points in 2016, and only slightly more likely to move to a large firm, which had been covered all along. We find no clear evidence that their wage growth slowed relative to men, including among workers who stayed with the same employer. These results suggest that small firms adjusted to the mandate through hiring and left the pay of women already on the payroll largely unchanged.

Third, the reform lowered the employment of mothers at small firms in the first years after birth. One year after the first birth, their employment fell by 1.8 percentage points relative to mothers at large firms, and the gap closed within about four years. Their earnings also fell in the first few years after birth and recovered later. In contrast, we find no decline in fathers' employment or earnings.

To interpret the firm response, we extend the wage-posting model of Burdett and Mortensen (1998), in which workers search on and off the job, to incorporate gender equal pay and job-protected leave. Taiwan's Gender Equality in Employment Act requires firms to pay women the same wage as men in the same job. In the model, when an employed woman gives birth, she takes job-protected leave, and her employer bears a cost to cover her absence, which falls with the number of coworkers who can take over her duties. As in Burdett and Mortensen (1998), a firm that pays more attracts and retains more workers who do not take leave, and each firm sets its wage by weighing this gain against the cost of paying more. The cost of leave does not enter this trade-off, so in the model no worker's wage falls. Gender equal pay then prevents the firm from paying women of childbearing age less than other workers. Unable to pass the cost of leave on through pay, a firm hires these women only if the rent it earns on a worker covers her expected leave cost. A firm with few employees faces a larger cost, because fewer coworkers can take over the absent worker's duties, and a firm that pays low wages earns a smaller rent on each worker. As a result, the model predicts that extending the obligation to small firms lowers their hiring of these women, most of all at the smallest and lowest-paying firms, without changing their wages relative to other workers. This is consistent with our findings that hiring fell at small firms, most strongly at the smallest ones, and that the wage growth of these women did not change relative to that of men of the same age.

To quantify the cost, we treat it as the unknown and back out the value that accounts for the observed hiring response. The calculation relies on two implications of the model. The job-finding rate of women of childbearing age relative to other workers pins down the share of firms that do not hire them, and each firm's rent can be recovered from the wage distribution of other workers (Bontemps et al., 2000). If every firm faced the same cost, the reform-induced decline in hiring would require a cost of about twice the annual minimum wage per year of employment. When the cost falls with the number of coworkers, the required cost drops to about one annual minimum wage, within the range of what disruptions to a small firm's work could plausibly cost, and the data cannot rule out a cost as small as the direct cost of covering an absence, roughly one day's wage paid to a coworker or a temporary replacement for each day of leave. The overall gap in job-finding rates between these women and other workers, which exists at firms of every size, implies a larger cost. This cost is an upper bound, because the gap may also reflect how actively these women look for work and other costs that employers expect after hiring them, such as

reduced working hours or exits after childbirth.

Our results have implications for current policy. Since January 2026, parents in Taiwan can take up to 30 days of leave one day at a time. The government also pays employers below the same 30-employee cutoff NT$1,000 (about US$31 at the 2015 exchange rate) for each such day of leave.[1] The payment is roughly one day of minimum-wage earnings, close to the direct cost of an absence. It may offset the cost the reform imposed on the smallest, lowest-paying firms, but it addresses only a small part of the cost that firms appear to attach to hiring women of childbearing age.

Related Literature

Our paper relates to work on how parental leave affects workers and how it affects firms. Most of this work examines one side of the market. We study both using the same policy variation and linked employer–employee records, and we find that the two sides respond in ways that reinforce each other.

Evidence on how leave expansions affect mothers is mixed. In Europe, longer paid leave has been found to raise female employment while lowering relative wages (Ruhm, 1998), to have little effect on earnings over the medium term (Schönberg and Ludsteck, 2014), and even to raise long run earnings for mothers with high earnings before birth (Frodermann et al., 2023). Studies from the United States reach less favorable conclusions. Mothers who took paid leave under California's program saw lasting declines in employment and earnings (Bailey et al., 2025), and exposure to state leave mandates before the Family and Medical Leave Act widened the motherhood penalty without affecting fathers (Flores et al., 2023). On the demand side, Timpe (2024) shows that the wage declines that followed the spread of temporary disability insurance as a source of paid maternity leave were concentrated in occupations where absence and turnover are most costly to employers.

A growing literature asks how firms absorb these costs, and the evidence remains unsettled. Leave extensions in Sweden and Denmark imposed meaningful costs on firms (Gallen, 2019; Ginja et al., 2023), and in Sweden the effects on the wage bill and on coworkers' hours were concentrated among firms below the median size (Ginja et al., 2023). In Austria, by contrast, Brenøe et al. (2025) find that firms adjust hiring, employment, and wages around a leave spell, return to baseline within a few years, and face no higher risk of closure. Whether employers can anticipate the absence appears to explain much of this

[1] Ministry of Labor, Guidelines for Rewarding Employers That Support Flexible Parental Leave (2025); Executive Yuan, press release on flexible parental leave, September 4, 2025. Government agencies, public schools, and state-owned enterprises are not eligible. Because the payment applies only to leave taken by the day, the costs it offsets may differ from those of the longer spells in our model.

disagreement. Where adverse effects do appear, they fall on the firms least able to spread an absent worker's tasks across others. Fernández-Kranz and Rodríguez-Planas (2021) show that after Spain gave parents of young children the right to a shorter workweek, employers became less likely to hire women of childbearing age, more likely to separate from them, and less likely to promote them into permanent contracts. Because nearly all users of the right were women, the law also penalized women who never had children, a pattern consistent with statistical discrimination. These effects were largest at establishments with fewer than ten employees.

Similar responses appear in other settings. In 2017, India extended fully paid maternity leave from 12 to 26 weeks for establishments with ten or more workers, with the cost borne entirely by employers. Using establishment data, Gupta (2023) finds that the female share of employment at covered establishments fell by one to two percentage points relative to smaller ones, while women's wages did not change significantly, and that the decline came mainly from the smaller covered establishments. When China lengthened statutory maternity leave in 2021, callback rates for young women on an online job platform fell by about 17 percent (Fang et al., 2025). In the United States, the Pregnancy Discrimination Act of 1978 reduced women's employment, consistent with firms hiring fewer women of childbearing age (Di Paolo and Marcolin, 2025), and mandated maternity leave has been linked to lower promotion rates for women, since employers who cannot tell which women will take leave treat all of them alike (Thomas, 2020). In these settings, employers typically bore part of the cost of the leave. Taiwan's reform differs in that paid leave was financed by employment insurance, so small firms bore the obligation to hold the job open but not the cost of the benefit.

Expected fertility can matter even when no mandate changes. Using the 2013 relaxation of China's One Child Policy, Sun (2026) finds that young women's wages fell by about 15 percent with no change in employment, mostly in jobs with long employment relationships. She attributes this to lower expectations of women's future productivity, and distinguishes it from a channel in which employers anticipate the cost of replacing a woman who leaves or takes leave (Amano-Patiño et al., 2020).

We add to this work in three ways. First, the 2009 reform required small firms to hold jobs open for the first time, while large firms had been required to do so all along, so large firms serve as a comparison group. Because paid leave was financed outside the firm, the only new cost the reform imposed on small firms was the obligation to hold a job open. Second, linked records let us observe the age and gender of each firm's workforce and new hires. We compare women of childbearing age with men of the same age, who are exposed to the same changes in the market for young workers but rarely take leave, and with older women, who are equally female but face little fertility risk. This comparison separates the response

specific to young women from a broader shift away from younger workers at small firms. Third, small firms adjusted through hiring while wages changed little. Our model focuses on the cost of covering a woman's absence, which is the channel Sun (2026) contrasts with her own explanation based on employers' expectations of women's future productivity. This cost can account for the response to the reform but is too small to account for the gap in job-finding rates that exists at firms of every size. Consistent with her findings, our results suggest that employers may expect costs that extend beyond the absence itself, and that whether they respond through wages or through hiring may depend on how much room they have to adjust pay, which is small at firms paying close to the minimum wage.

The concentration of these effects at small firms connects our paper to the literature on regulation tied to firm size. Many labor laws apply only above an employment threshold, and the resulting discontinuities have been used to measure the cost of regulation, for example the obligations French firms face at 50 employees (Garicano et al., 2016) and the stricter dismissal rules Italian firms face at 15 (Schivardi and Torrini, 2008). Most of these studies compare firms just above and just below a fixed cutoff, which raises concerns that firms sort around the threshold and that departures of the firm size distribution from a Pareto shape are attributed to the regulation itself (Gourio and Roys, 2014). Our design instead uses a change in the threshold. By extending leave with job protection from firms with 30 or more employees to all firms, the 2009 reform turned previously exempt firms into covered ones, in the spirit of Bauernschuster (2013), who exploits a shift in the size threshold of German dismissal protection, and Morikawa (2022), who studies a Japanese reform that raised several regulatory thresholds.

The remainder of the paper is organized as follows. Section 2 describes the institutional setting and the administrative data. Section 3 presents the reduced-form evidence on firm and worker responses to the 2009 reform. Section 4 extends the Burdett–Mortensen model to incorporate gender equal pay and job-protected leave and uses it to back out the cost of leave that accounts for the hiring response. Section 5 concludes.

2 Institutional Background and Data

This section describes the institutional setting and the administrative data used in our analysis. We first outline Taiwan's parental leave system and the 2009 reform, which removed the earlier 30-employee firm-size restriction on job-protected leave and introduced government-funded leave benefits. We then introduce the nationwide administrative dataset from the Fiscal Information Agency of Taiwan, covering 2006 to 2017, and describe the sample construction underlying our difference-in-differences design and

worker-level event study analysis.

2.1 Institutional Policy Setting

Taiwan, like many East Asian economies, has experienced a sharp and persistent decline in fertility over the past two decades. The total fertility rate fell to 1.1 by 2008, among the lowest in the world at the time,[2] even as female labor force participation has risen. Concerned that the conflict between employment and childrearing was contributing to this decline, the Taiwanese government has progressively expanded family-friendly labor market policies, including job-protected and paid parental leave, which is the focus of this paper.

Unpaid, job-protected parental leave was first introduced in Taiwan in 2002 under the Gender Equality in Employment Act, with the explicit goal of improving gender equality in the labor market. Under the original law, only employees at firms with 30 or more employees were eligible. Eligible workers could take up to two years of leave, provided their child was under three years of age and they had been employed at the firm for at least one year.[3] Because eligibility depended entirely on firm size, independent of individual or job characteristics, the 2002 law effectively created two classes of workers, those employed at firms with 30 or more employees, who enjoyed both job protection and the right to return to their pre-leave position, and those at smaller firms, who had no entitlement to leave at all and risked losing their job if they took time off to care for a newborn.

This firm-size restriction was removed in 2009. Under the amended law, all employees became eligible for job-protected parental leave regardless of firm size. The 2009 amendment simultaneously introduced a government-funded parental leave allowance, financed through the employment insurance fund, which we refer to as paid leave.[4] Eligible employees are entitled to a leave allowance equal to 60% of their average insured salary over the six months prior to taking leave. To qualify, an employee must have been enrolled in the employment insurance system for at least one year prior to taking leave. The paid allowance is disbursed monthly and can be received for up to six months per parent per child; if both parents are eligible, they may each claim up to six months, for a combined maximum of twelve months of paid benefits per child. Job-protected leave can last longer than the paid benefit. It can be taken for up to two years per child, regardless of how the leave is divided between parents, must be completed before the child turns three, and is unpaid beyond the period covered by the allowance. Prior to a further

[2]Source: Ministry of the Interior, Department of Household Registration, Taiwan.

[3]During the leave period, employees remain eligible to participate in the social security system.

[4]The employment insurance fund is supported by contributions from employees, employers, and the government, so the cost of the cash benefit is not borne by the firm.

amendment in 2022, parents could not take this leave simultaneously, so if both parents took leave for the same child, they did so consecutively within that two-year cap. Because the allowance is tied to the insured salary, which is itself subject to a statutory ceiling, the benefit is capped in practice. In 2009, the insured monthly salary ceiling was 43,900 NTD (approximately USD 1,330), implying a maximum monthly benefit of 26,340 NTD (approximately USD 800).[5]

Because eligibility for the pre-2009 regime depended solely on firm size, the reform changed different things for workers at small and large firms. Workers at firms with 30 or more employees already had job-protected leave before 2009 and gained paid leave. Workers at firms with fewer than 30 employees gained both job-protected leave and paid leave. Table 1 summarizes this exposure by firm size and reform period, and Appendix Table 9 presents the full policy detail.

Table 1: Policy Exposure by Firm Size and Period

		Pre-reform	Post-reform	
		Mar 2002 – Apr 2009	May 2009 –	Change at reform
Small firms	Job-protected leave	no	yes	Newly granted
(fewer than 30 employees)	Leave allowance	no	yes	Newly granted
Large firms	Job-protected leave	yes	yes	Unchanged
(30 or more employees)	Leave allowance	no	yes	Newly granted

Notes: Both groups gain the employment-insurance allowance in May 2009; only small firms additionally gain the right to job-protected leave. Therefore, the difference-in-differences contrast identifies the job-protection component, plus any differential response to the allowance across firm sizes. Full policy detail is presented in Appendix Table 9.

Figure 1 plots parental leave take-up rates by gender from 2002 to 2020, using external administrative statistics from the Ministry of Labor. Take-up rose sharply among women after the 2009 reform, providing evidence that the policy change induced a meaningful behavioral response. Take-up among men rose only slightly over the post-policy period, which motivates our use of men of the same age as a comparison group in the analysis that follows. The two segments of the series measure different underlying events. The dashed line through 2008 counts unpaid parental leave applications,[6] while the solid line from 2009 onward counts first approvals of the new paid leave allowance.[7] After 2009, workers can extend their leave beyond the paid period as unpaid, job-protected leave, but the published statistics do not report this unpaid portion separately, so we cannot show a separate series for it.Because the composition of the

[5]Converted at the 2009 average exchange rate of 33.02 NTD/USD (Federal Reserve Board, G.5A).

[6]Filed under the Gender Equality in Employment Act, which took effect on 8 March 2002.

[7]Introduced on 1 May 2009 through the Employment Insurance fund.

numerator changes at this cutoff, the pre- and post-2009 series are not directly comparable in level, and the figure should be read as evidence of two distinct regimes. Take-up of unpaid leave before 2009 was low, which may also help explain why firms did not bunch below the 30-employee threshold.

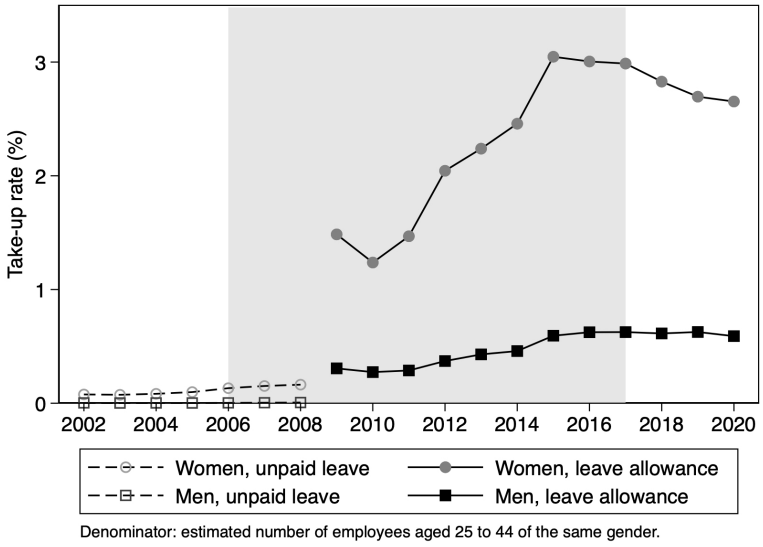


Figure 1: Parental leave take-up rate by gender, 2002–2020

Notes: Take-up rate is the number of parental leave cases divided by the estimated number of employees aged 25 to 44 of the same gender. Figures cover Labor and Employment Insurance only and exclude the public-service and military schemes. 2002 covers ten months, since the Act took effect on 8 March 2002, and 2009 covers eight months; both are scaled to a full-year basis. Denominators are employed persons aged 25 to 44 by gender, scaled by the gender-specific share of employees in total employment, since the allowance is available only to insured employees and no employee count by age and gender is published. *Sources:* Ministry of Labor, *Statistical Yearbook of Labor Trends* (2010 ed., p. 30) for 2002–2008; Ministry of Labor, *Gender Labor Statistics* (2020 ed., Table 5-7) for 2009–2020; Ministry of Labor, *Gender Labor Statistics*, Tables 3-1 and 3-5, based on the DGBAS Manpower Survey, for the denominators.

2.2 Data

We use several nationwide administrative datasets from the Fiscal Information Agency (FIA), of the Ministry of Finance in Taiwan, covering the entire population of Taiwan from 2006 to 2017. The dataset is constructed from mandatory tax filing records, giving it near-universal coverage of formal-sector employment relationships over this period, and allows us to track individual workers and firms.[8]

[8]Because the data are drawn from administrative records, they are not subject to sampling error or self-reported measurement error in earnings and employment.

Worker-level data are primarily from individual income tax records, which provide information on employers and earnings. An individual is defined as employed in a year if she has positive earnings in that year, and earnings are calculated by adding the pay from employers and the withheld labor pension for each year. For individuals who earn earnings from multiple employers in a given year, total earnings are aggregated and the employer providing the highest earnings is assigned as the primary employer. We measure firm size as the number of workers for whom the firm is the highest-paying employer in a given year, counting part-time and full-time workers equally. This headcount measure follows the way statutory firm-size thresholds are applied in Taiwan.[9] All earnings are deflated using the Consumer Price Index (CPI), with 2015 as the base year. An individual-household dataset is constructed by linking tax registry records to individual income tax records, which provide demographic information, including birth year and gender, and allow household relationships to be recovered and couples and their children to be identified. Since our interest is in the effect of first childbirth, we focus on the year of first birth and do not discuss other parities in this paper. Table 2 presents the summary statistics at the worker level.

Firm-level data are derived from firms' business tax and income tax records, which contain detailed financial statements such as income statements, balance sheets, cost of revenue, and manufacturing expense schedules. These records provide information on employment, revenue, costs, and expenses. Merging these data with the business entity registry yields supplementary firm-level information, including ownership type (private or public), registered location, and industry codes. Linking the individual-household dataset to the firm-level dataset yields a linked employer-employee dataset, from which the composition of the workforce within each firm is computed, including the share of female employees, the proportion of women of childbearing age, and the overall age distribution. Firm-level characteristics are summarized in Table 3.

The public administration and defense sector is excluded because public employees are covered under a separate insurance system with its own parental leave regulations that are unaffected by the 2009 amendment we study. Including these workers would contaminate both our treatment and comparison groups with individuals whose leave entitlements did not change.[10] For the same reason, we also exclude school employees[11], since these sub-sectors consist overwhelmingly of public schools whose teaching staff

[9] Taiwan's Gender Equality in Employment Act and Labor Standards Act define an employee as any person employed by an employer to work for wages, without distinguishing full-time from part-time work, and neither statute converts part-time workers into full-time equivalents. Therefore, thresholds are applied on a headcount basis.

[10] Public employees are enrolled in the Insurance for Government Employees and School Staff. Their parental leave is governed by separate personnel regulations, and the corresponding leave allowance took effect on August 1, 2009 under a separate amendment, on a different schedule and without any firm-size threshold.

[11] This includes employees in preschool, elementary, junior secondary, senior secondary, and special education. Because our

are covered under the same separate insurance system. Firms in agriculture, forestry, fishing, and animal husbandry (industry major group A) are also excluded, because firms in this sector differ systematically in size, seasonality of employment, and reporting practices in ways that make firm size a less reliable proxy for regulatory treatment status.[12]

The estimation sample is constructed at two levels. At the firm level, the pre-reform (2007) employee headcount is computed for every firm in the sample. Following Ginja et al. (2023), firms with fewer than three employees and those in the top 1% of the 2007 firm-size distribution (more than 264 employees) are excluded to limit the influence of very small and very large firms. The remaining firms are classified as treated (fewer than 30 employees) or comparison (30 or more employees) based on this fixed baseline. At the worker level, individuals employed at treated and comparison firms are identified and followed over time to construct event-study samples around childbirth, allowing the dynamic effects of the reform on employment continuity and earnings to be traced out.

To assess whether the reform altered the shape of the earnings distribution, we compare empirical CDFs of real annual earnings before and after the reform, separately by gender, firm size, and age group (childbearing age 20–39, non-childbearing age 40–59, and ages 25–34), reported in Appendix Figure 11. Within each subgroup, the pre- and post-policy CDFs are visually similar. Appendix Table 10 formalizes this comparison using two-sample Kolmogorov-Smirnov tests. While all comparisons formally reject the null of equal distributions given the large sample size, the estimated D statistics range from 0.015 to 0.123, and the divergence between the pre- and post-policy distributions is modest across all subgroups.

3 Empirical Evidence

As described in Section 2, the 2009 parental leave policy extended parental leave eligibility to all workers regardless of firm size. In this section, we first examine how firms responded to the policy change, in whom they hired and employed, and then turn to the mobility and pay of childbearing-age women and to parents' labor market outcomes following childbirth. Throughout our analysis, we exclude 2009 from the sample. Because the reform took effect on May 1, 2009, individuals observed in 2009 experienced only

data do not permit us to distinguish public from private employees within these industries, we exclude them in their entirety to avoid contaminating the sample.

[12] In agriculture, forestry, fishing, and animal husbandry, self-employed farmers and fishers are enrolled in Farmers' Health Insurance, and Article 5 of the Employment Insurance Act excludes employees of employers that are not required to register. The leave entitlement under the Gender Equality in Employment Act applies to all industries, so these exclusions follow from coverage by the insurance program that pays the benefit, not eligibility for leave itself.

Table 2: Worker-Level Summary Statistics at Baseline ($s = -1$)

	Pre-reform cohort						Post-reform cohort					
	Small		Large				Small		Large			
Variable	Mean	SD	Mean	SD	Diff.	Norm. Diff.	Mean	SD	Mean	SD	Diff.	Norm. Diff.
Panel A: Mothers												
Age	30.822	7.550	30.349	6.314	0.473	0.068	31.217	6.072	31.337	5.243	−0.120	−0.021
Employment	1	0	1	0	0	0	1	0	1	0	0	0
Earnings (1000s USD)	9.007	6.030	12.077	8.069	−3.070	−0.431	9.279	6.093	13.099	8.206	−3.820	−0.529
Employed in manufacturing	0.233	0.423	0.354	0.478	−0.121	−0.268	0.186	0.389	0.292	0.455	−0.106	−0.250
Employed in Services	0.376	0.484	0.208	0.406	0.168	0.376	0.474	0.499	0.348	0.476	0.126	0.258
N	26,834		29,600				105,174		116,027			
Panel B: Fathers												
Age	32.787	7.737	32.574	6.761	0.213	0.029	32.940	6.225	33.040	5.650	−0.100	−0.017
Employment	1	0	1	0	0	0	1	0	1	0	0	0
Earnings (1000s USD)	11.371	8.091	17.352	11.153	−5.981	−0.614	11.511	8.320	17.366	10.839	−5.855	−0.606
Employed in manufacturing	0.334	0.472	0.419	0.493	−0.085	−0.176	0.289	0.453	0.380	0.485	−0.091	−0.194
Employed in Services	0.448	0.497	0.234	0.423	0.214	0.464	0.467	0.499	0.311	0.463	0.156	0.324
N	31,121		33,236				104,393		122,511			

Notes: Sample includes workers employed in the year before childbirth ($s = -1$). Small (large) firms are those whose employer in the year before birth had fewer than (at least) 30 employees in 2007. The pre-reform cohort includes births before 2009, and the post-reform cohort includes births after 2009. Diff. is the small-minus-large difference in means within each period. Norm. Diff. is the normalized difference following Imbens and Rubin (2015), computed as $(\bar{X}_S - \bar{X}_L)/\sqrt{(s_S^2 + s_L^2)/2}$, where $\bar{X}_S$ and $\bar{X}_L$ are the means and s_S and s_L the standard deviations for small and large firms. Educational attainment is not available in the FIA administrative data, so we omit it from this table. Earnings are converted from NTD to USD using the 2007 annual average exchange rate of NTD/USD = 32.842, the same year as the baseline used to define firm size.

Table 3: Firm-Level Summary Statistics (2007 Baseline)

	Small Firms		Large Firms			
Variable	Mean	SD	Mean	SD	Diff.	Norm. Diff.
Firm size						
Number of employees	8.344	6.017	71.118	49.171	−62.774	−1.792
Gender composition						
Female share	0.472	0.255	0.478	0.200	−0.005	−0.022
Childbearing-age female share						
Age 18–49	0.339	0.255	0.356	0.196	−0.017	−0.075
Age 20–39	0.220	0.232	0.241	0.167	−0.020	−0.100
Age 25–34	0.121	0.168	0.136	0.112	−0.015	−0.105
Non-childbearing-age female share						
Age 40–59	0.190	0.195	0.162	0.133	0.028	0.167
Industry composition						
Manufacturing share	0.219	0.413	0.336	0.472	−0.118	−0.265
Services share	0.479	0.500	0.336	0.472	0.143	0.293
Annual earnings (1000s USD)						
Average earnings per employee	7.445	5.273	10.571	8.160	−3.125	−0.455
Average earnings per employee (above min. wage)	11.427	6.248	16.073	8.783	−4.646	−0.610
Share of new hires						
Female share	0.466	0.373	0.470	0.235	−0.003	−0.011
Childbearing-age female share						
Age 18–49	0.343	0.363	0.345	0.237	−0.002	−0.007
Age 20–39	0.251	0.331	0.254	0.211	−0.002	−0.009
Age 25–34	0.132	0.249	0.137	0.146	−0.004	−0.021
N (firms)	326,823		38,126			

Notes: Small firms have fewer than 30 employees and large firms have 30 or more employees, measured in 2007 (pre-reform baseline). The sample excludes public administration/defense and school employees, both covered under a separate insurance system with unaffected parental leave rules (schools cannot be split into public/private in the data, so are dropped entirely), and agriculture/forestry/fishing/animal husbandry, where firm size is a less reliable proxy for treatment status. Average annual earnings are converted from NTD to USD using the 2007 annual average exchange rate of NTD/USD = 32.842, the same year as the baseline used to define firm size. Diff. is the small-minus-large difference in means. Norm. Diff. is the normalized difference following Imbens and Rubin (2015), computed as $(\bar{X}_S - \bar{X}_L)/\sqrt{(s_S^2 + s_L^2)/2}$, where $\bar{X}_S$ and $\bar{X}_L$ are the means and s_S and s_L the standard deviations for small and large firms.

partial exposure to the policy within that calendar year, making it difficult to classify 2009 as either a clean pre- or post-reform period. As shown in Appendix Figure 9, the firm size distribution exhibits no discernible mass point at or just below the 30-employee threshold relative to nearby firm-size values. We find no evidence that treatment assignment based on 2007 firm size was driven by manipulation.

Two features of the timing deserve attention. First, the removal of the firm-size threshold was promulgated in January 2008, more than a year before it took effect (Appendix Table 9), so firms may have adjusted in anticipation during 2008, our reference year. Anticipation would lower the 2008 hiring share of childbearing-age women at small firms and make the 2007 coefficient positive. The estimated 2007 coefficient for the hiring share is instead negative, so we find little evidence of anticipation on the hiring margin. Second, the reform coincided with the 2008–09 global financial crisis, during which unpaid furloughs were widespread in Taiwan, particularly in manufacturing. Small and large firms may have been hit by the crisis and recovered from it differently. Industry-by-year fixed effects absorb shocks common to all firms in an industry, and the triple-difference estimates, which compare childbearing-age women with men of the same age at the same firms, absorb shocks that affected the hiring of young workers at small firms regardless of sex.

Each part of the analysis addresses a specific question. We start with the hiring share, the margin that firms control most directly, and then turn to the composition of the workforce, which shows whether the change in hiring accumulated over time. Pooled estimates summarize these responses, compare them with those for men of the same age, and show how they vary with firm size, which the model in Section 4 uses to discipline how the cost of leave varies across firms. We then ask what happened to the women themselves: whether they moved to other employers, whether their wages changed relative to men, which the model predicts they should not, and how the reform affected the employment and earnings of mothers after childbirth.

3.1 Employer Responses

In this section, we examine whether the reform changed small firms' hiring decision and workforce composition. The reform raised the costs small firms face when an employee gives birth. They must now hold her position open during leave, cover her work or find a temporary replacement, and bear the uncertainty of whether she will return. Because these costs arise only when an employee gives birth, firms concerned about them have an incentive to reduce their exposure to workers most likely to have children. We focus on women of childbearing age, the group whose expected leave-taking is highest, and compare them with women outside childbearing age, who are unlikely to take leave but otherwise face

similar labor market conditions. If small firms shifted their workforce away from childbearing-age women after the reform, relative to both large firms and non-childbearing-age women, this would point to firms treating childbearing risk as a cost of employment and adjusting their hiring accordingly.

The first question we ask is how small firms changed whom they hire. When a firm fills a position, it has to form expectations about the new worker's future costs, including how likely the worker is to take leave. If small firms internalized the cost of the reform, this should show up in their hiring decisions and, over time, in the composition of their workforce. We ask whether small firms became less likely to fill new positions with women of childbearing age after 2009 and whether this changed who worked at these firms. To trace the timing of this response, we estimate a dynamic difference-in-differences specification that compares small firms, which became newly covered in 2009, with large firms, which were covered throughout,

$$Y_{jt} = \alpha \, \text{Small}_j + \sum_{k \notin \{2008,\, 2009\}} \beta_k \left(\text{Small}_j \times \mathbb{1}[t = k] \right) + \xi_{n(j)t} + \varepsilon_{jt}, \tag{1}$$

where Small_j equals one if firm j had fewer than 30 employees in 2007 and became newly covered by the parental leave mandate in 2009, $\xi_{n(j)t}$ are industry-by-year fixed effects, which also absorb year and industry effects, and ε_{jt} is the error term. Standard errors are clustered at the firm level.

We use two outcome variables, the hiring share and composition. The hiring share is the share of firm j's new hires in year t who belong to a given group, where a new hire is a worker employed at firm j in year t but not in year $t-1$. Composition is the share of firm j's employees in year t who belong to a given group. For each measure, we consider childbearing-age women (20 to 39), men of the same age, older women (40 to 59), and older men. We first estimate equation (1) separately for each group. To remove changes common to an age group, we also estimate equation (1) with the gap between the shares of women and men of the same age as the outcome,

$$Y_{jt}^{\text{gap},a} = Y_{jt}^{W,a} - Y_{jt}^{M,a}, \qquad a \in \{20\text{–}39,\ 40\text{–}59\}, \tag{2}$$

where $Y_{jt}^{W,a}$ and $Y_{jt}^{M,a}$ are the shares of women and men in age group a among firm j's new hires (or employees) in year t. We refer to the resulting estimates as triple-difference estimates.[13]

Because our data record annual earnings, a woman whose parental leave covers an entire calendar year has no wage income from her employer in that year and drops out of the firm's workforce, and she is counted as a new hire when she returns. After the reform, this mechanically lowers the composition share

[13] Each triple-difference estimate and its standard error come from a single regression, so the standard error accounts for the correlation between the two shares within a firm. Because the regressions for women and men use the same sample and regressors, the point estimate equals the difference between their coefficients.

of childbearing-age women at small firms and raises their hiring share. This bias works against finding a decline in the hiring share and may account for part of the decline in composition. Since the average leave lasts about seven months, absences spanning a full calendar year are uncommon, and we expect this channel to be small.

$\mathbb{1}[t = k]$ is an indicator for year k. The reference year is 2008, and 2009 is excluded because the reform took effect in the middle of that year. For composition, k runs over 2006, 2007, and 2010–2017. Because new hires are defined relative to the previous year, 2007 is the earliest year in which the hiring share is observed, so for the hiring share k runs over 2007 and 2010–2017. The coefficients of interest, β_k, trace out the difference in the outcome between small and large firms in year k, relative to the same difference in 2008. Under parallel trends, the pre-reform coefficients should be zero.

3.1.1 Hiring

Figure 2 plots the estimated β_k for the share of new hires in each gender and age group. Before turning to the post-reform estimates, we examine the pre-reform coefficient. For childbearing-age women, the 2007 coefficient is -0.8 percentage points and statistically different from zero. Relative to large firms, the share of childbearing-age women among new hires at small firms was rising slightly before the reform, which is the opposite direction of the post-reform effect. Men of the same age have a similar pre-reform coefficient (-1.0 percentage points), which suggests that it reflects age-specific hiring patterns at small firms that apply to men and women alike. Because the pre-reform coefficient is not zero, we examine the sensitivity of our estimates to violations of parallel trends using the approach of Rambachan and Roth (2023), and we also estimate a triple-difference specification that compares women with men of the same age.

Panel (A) of Figure 2 shows that after the reform, the share of childbearing-age women among new hires at small firms fell relative to large firms. The decline is about 0.8 percentage points in 2010 and 2011, grows to 1.3 percentage points in 2012, and reaches about 2 percentage points by 2016 and 2017. The other panels show how the composition of new hires adjusted. The share of older women among new hires rose steadily, from 0.4 percentage points in 2011 to about 1.3 percentage points by 2016. For same-age men, the coefficients are small and mostly insignificant through 2014 and turn negative afterward, reaching -1.1 percentage points in 2017. The coefficients for older men remain small throughout, at most 0.5 percentage points.

Because the 2007 coefficient is not zero, part of the post-reform decline could continue a trend that started before the reform. The approach of Rambachan and Roth (2023) asks how large such violations of parallel trends would have to be to overturn our conclusions, by bounding post-reform deviations from

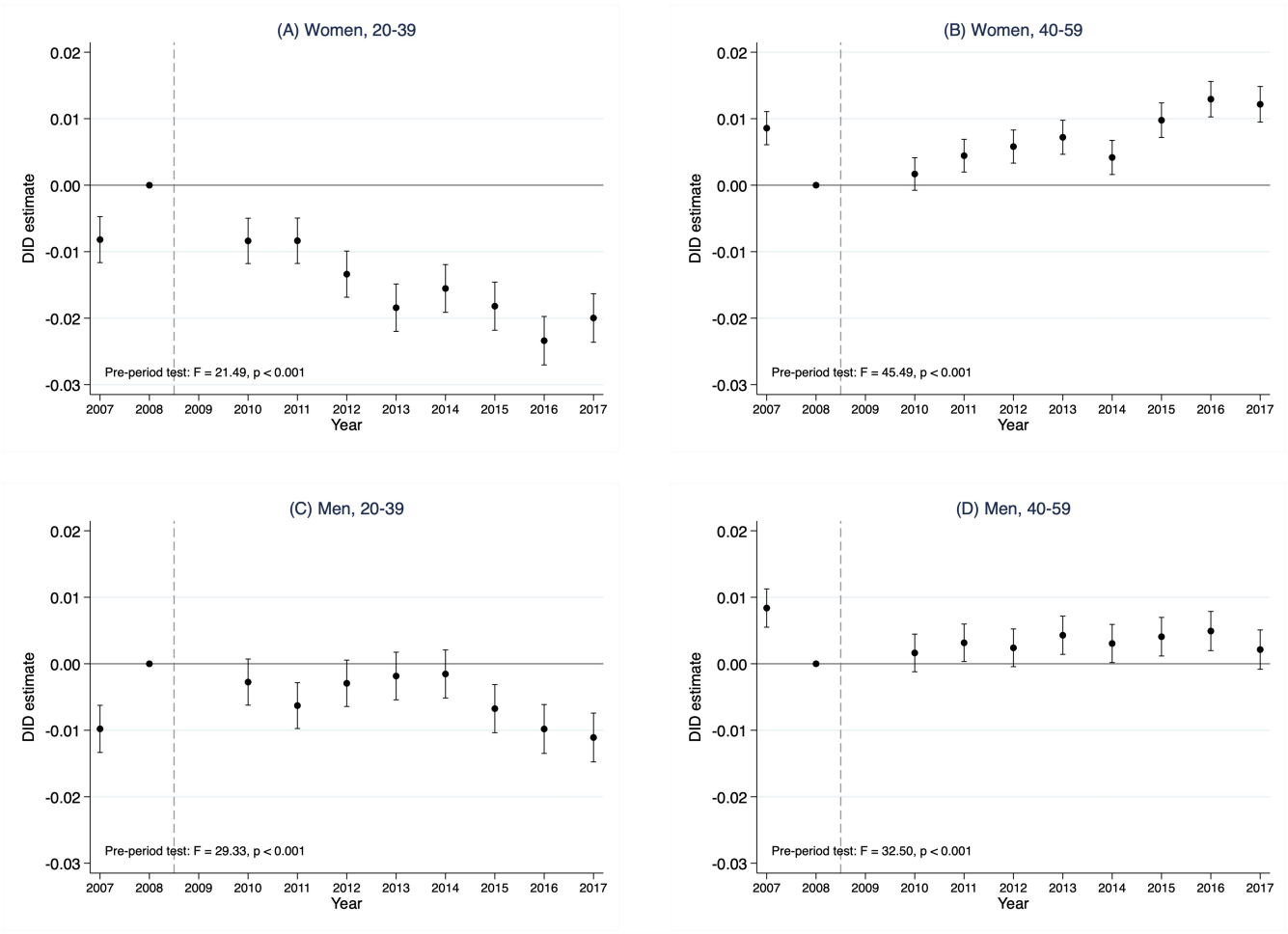


Figure 2: Event-Study Estimates: Hiring Share by Gender and Age

Note: This figure plots the estimated β_k for the share of new hires in each gender and age group. Small firms are those with fewer than 30 employees in 2007, which became newly subject to the parental leave mandate in 2009. The reference year is 2008; 2009 is excluded because the reform took effect in the middle of that year. All regressions include industry $\times$ year fixed effects, and standard errors are clustered at the firm level. Bars show 95% confidence intervals. The pre-period test reports the F-statistic and p-value for the null that the 2007 coefficient is zero.

parallel trends relative to the deviations observed before the reform. Table 4 reports the results. Under relative magnitudes, post-reform violations of parallel trends are bounded by $\bar{M}$ times the largest pre-reform change. For the hiring share, the robust confidence set for the average post-reform effect excludes zero for values of $\bar{M}$ up to 0.3, and the 2010 effect remains significant for $\bar{M}$ up to 0.7. Since the largest pre-reform change is 0.8 percentage points, $\bar{M} = 0.3$ already allows for a cumulative bias of about 1.1 percentage points in the average effect, roughly 70 percent of the point estimate. If the bias is restricted to have the same sign as the pre-reform change, the confidence set excludes zero for all values of $\bar{M}$ up to 1.5. Because this restriction rests on a single pre-reform coefficient, we treat it as supplementary evidence. Restricting the bias to the direction that would overstate the decline leaves the breakdown values unchanged. Under the smoothness restriction with $M = 0$, which extends the pre-reform trend linearly, both the 2010 and the average effects remain significant.

Table 4: Sensitivity to Violations of Parallel Trends

	Hiring share Figure 2 (A)		Composition Figure 4 (A)	
	2010	Average	2010	Average
Relative magnitudes, $\bar{M}$	0.7	0.3	0.8	0.35
with sign restriction: same sign as pre-reform change	> 1.5	> 1.5	0.8	0.35
with sign restriction: bias overstates the decline	0.7	0.3	0.8	0.35
Smoothness, $M = 0$	Significant	Significant	Significant	Significant

Notes: The table reports results from the sensitivity analysis of Rambachan and Roth (2023). Each entry is the largest value of $\bar{M}$ at which the 95 percent robust confidence interval excludes zero. The outcomes are the share of childbearing-age women among new hires (hiring share) and among employees (composition). 2010 is the first post-reform coefficient, and Average is the average of the 2010 to 2017 coefficients. Under relative magnitudes, each post-reform year-to-year violation of parallel trends is at most $\bar{M}$ times the largest pre-reform year-to-year change. The first sign restriction requires the post-reform violation to have the same sign as the pre-reform change; for the hiring share, this relies on a single pre-reform coefficient. The second requires the violation to be negative, the direction that would overstate the estimated decline; for composition, the two restrictions coincide. Under smoothness, the slope of the differential trend may change by at most M per year; $M = 0$ extrapolates the pre-reform linear trend. 2009 is excluded, so the step from 2008 to 2010 is treated as one period.

Using the difference between the shares of women and men of the same age as the outcome removes changes in hiring that are common to an age group. Figure 3 reports the resulting estimates of β_k. In Panel A, the outcome is the hiring share of childbearing-age women minus that of men of the same age.

The 2007 coefficient is small and statistically insignificant, so we find no evidence of a pre-trend in this gap. After the reform, the gap at small firms declines. The estimates for 2010 and 2011 are negative but imprecise. From 2012 on, they are larger and statistically significant, reaching −1.7 percentage points in 2013 and ranging between −0.9 and −1.4 percentage points through 2017. After the reform, small firms filled a smaller share of their new positions with women of childbearing age than with men of the same age.

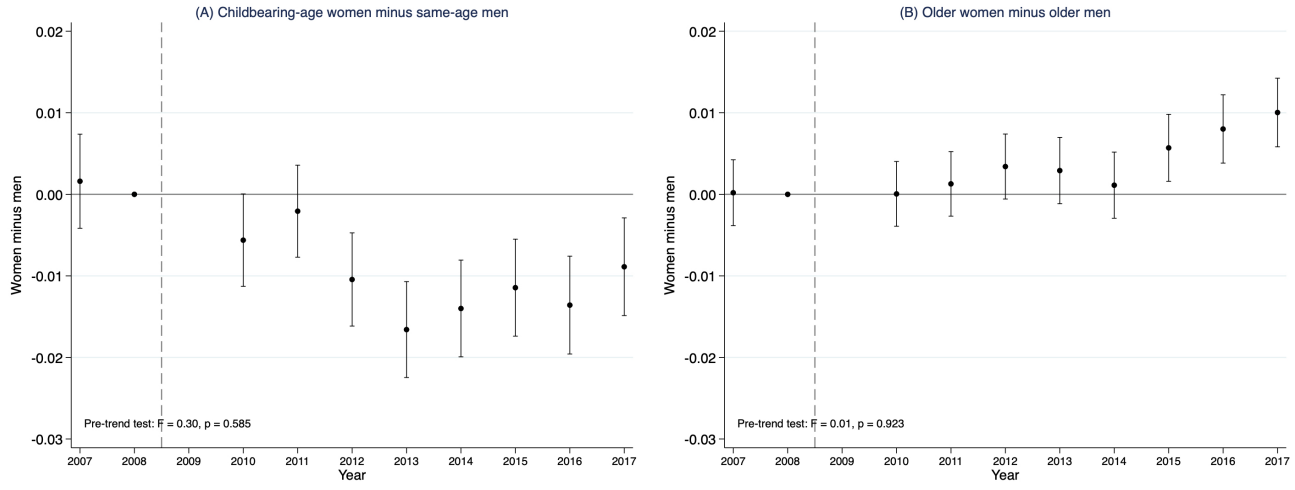


Figure 3: Triple-Difference Estimates: Hiring Share, Women Minus Men of the Same Age

Note: This figure plots the estimated β_k from regressions in which the outcome is the firm-level difference between the share of women and the share of men of the same age among new hires. Panel A compares childbearing-age women (20 to 39) with men of the same age, and Panel B compares older women (40 to 59) with older men. The reference year is 2008; 2009 is excluded. All regressions include industry × year fixed effects, and standard errors are clustered at the firm level. Bars show 95% confidence intervals. The pre-trend test reports the F-statistic and p-value for the null that the 2007 coefficient is zero.

Panel B repeats the exercise for older women and older men. Again, the 2007 coefficient is close to zero and statistically insignificant. The post-reform estimates are small and insignificant through 2014 and rise to 1.0 percentage point by 2017. Small firms hired relatively fewer women of childbearing age and, in later years, relatively more older women. A general reluctance to hire women would push both gaps down. The opposite movements in the two panels instead suggest that small firms substituted toward women who were unlikely to take leave.

3.1.2 Workforce Composition

The hiring share captures the margin that firms control most directly. We next ask whether the change in hiring accumulated into a different workforce at small firms. Composition responds more slowly and

also reflects separations, the aging of incumbent workers, and, as noted above, full-year leave spells that remove women from the annual records, so we treat it as supporting evidence. We repeat the analysis for the composition of employees. Figure 4 shows that the pre-reform coefficients for childbearing-age women decline from 1.8 percentage points in 2006 to 0.3 percentage points in 2007 and are jointly statistically different from zero. Unlike the hiring share, this pre-reform movement runs in the same direction as the post-reform effect. The pattern is not specific to women. The share of same-age men also declines before the reform, while the shares of older women and older men rise, as would happen if the workforce at small firms were gradually aging. Next, we apply the Rambachan and Roth (2023) approach and the triple-difference specification.

After the reform, the share of childbearing-age women among employees at small firms falls by 1.5 percentage points in 2010 and by about 3.2 percentage points by 2014, where it remains through 2017. The share of older women rises by a similar magnitude, from 1.1 percentage points in 2010 to 2.3 percentage points in 2017. The share of same-age men declines by 1.1 to 1.7 percentage points, and the share of older men rises until 2012 and falls below zero by 2016.

Because the pre-reform coefficients for composition follow a trend, the smoothness restriction of Rambachan and Roth (2023) provides the most informative benchmark (Table 4). With $M = 0$, which extrapolates the pre-reform trend linearly, the average post-reform effect remains negative and significant, with a robust confidence set of $[-2.4, -0.8]$ percentage points. This result is sensitive to allowing the trend to bend, but the 2010 effect remains significant for all values of M we consider. Under relative magnitudes, the average effect is robust for $\bar{M}$ up to 0.35 and the 2010 effect for $\bar{M}$ up to 0.8. Since the pre-reform change and the estimated effect have the same sign, restricting the bias to that direction does not change these values.

Figure 5 presents the triple-difference estimates for composition. In Panel (A), the 2007 coefficient is essentially zero (-0.06 percentage points), while the 2006 coefficient is 1.1 percentage points, so the two pre-reform coefficients are still jointly significant. Compared with the difference-in-differences estimates, the pre-reform movement is smaller and confined to 2006. After the reform, the gap between childbearing-age women and same-age men declines by 0.4 percentage points in 2010, widens to about 1.6 to 1.7 percentage points in 2013 and 2014, and stays near 1.5 percentage points through 2017. All post-reform coefficients are statistically significant. In Panel (B), the gap between older women and older men shows no significant pre-reform difference and rises steadily after the reform, from 0.5 percentage points in 2010 to 2.7 percentage points in 2017.

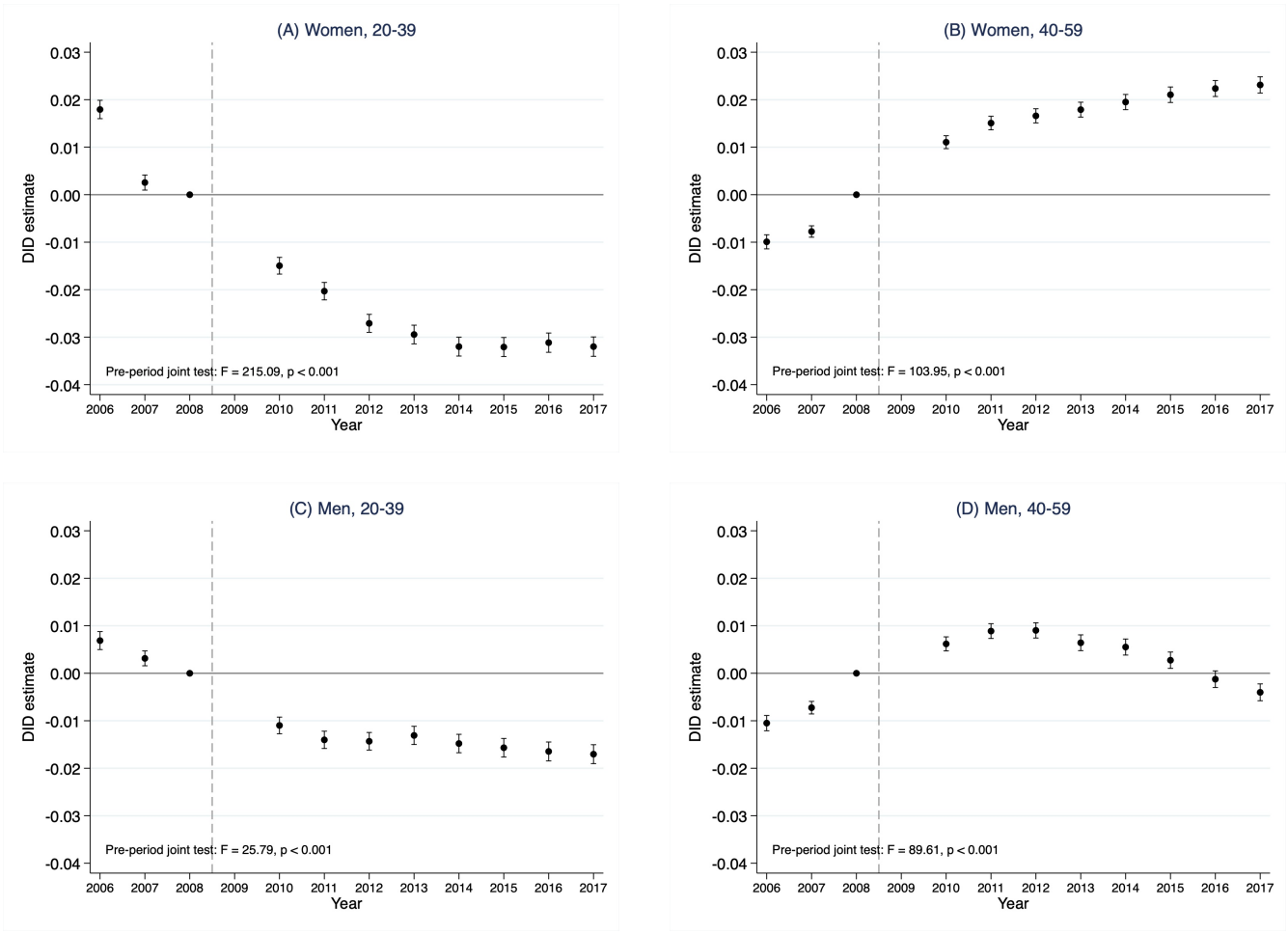


Figure 4: Event-Study Estimates: Workforce Composition by Gender and Age

Note: This figure plots the estimated β_k for the share of employees in each gender and age group. Small firms are those with fewer than 30 employees in 2007, which became newly subject to the parental leave mandate in 2009. The reference year is 2008; 2009 is excluded because the reform took effect in the middle of that year. All regressions include industry $\times$ year fixed effects, and standard errors are clustered at the firm level. Bars show 95% confidence intervals. The pre-period joint test reports the F-statistic and p-value for the null that the 2006 and 2007 coefficients are jointly zero.

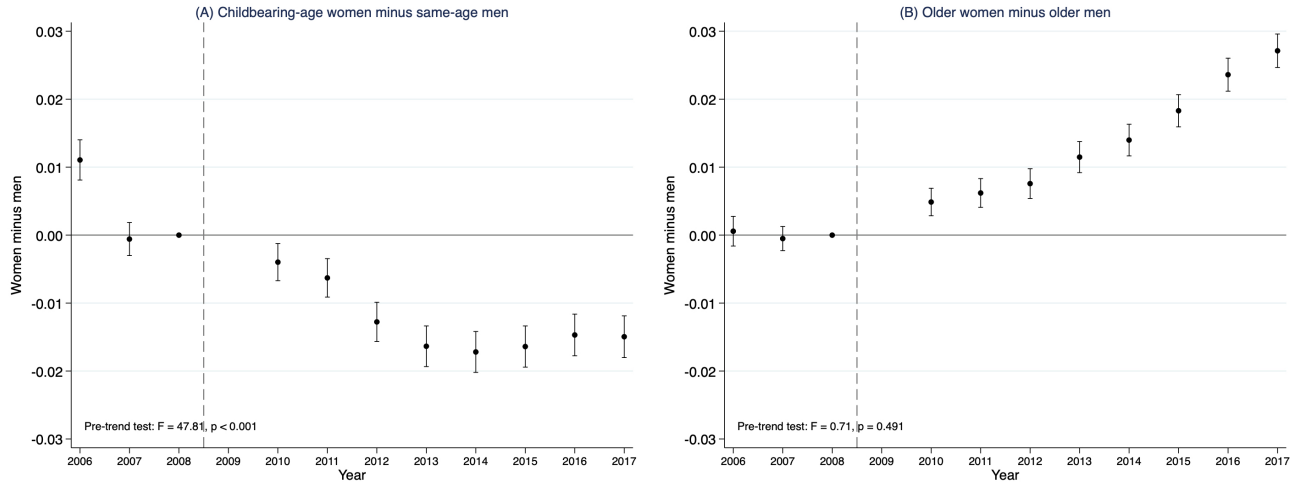


Figure 5: Triple-Difference Estimates: Workforce Composition, Women Minus Men of the Same Age

Note: This figure plots the estimated β_k from regressions in which the outcome is the firm-level difference between the share of women and the share of men of the same age among employees. Panel A compares childbearing-age women (20 to 39) with men of the same age, and Panel B compares older women (40 to 59) with older men. The reference year is 2008; 2009 is excluded. All regressions include industry $\times$ year fixed effects, and standard errors are clustered at the firm level. Bars show 95% confidence intervals. The pre-trend test reports the F-statistic and p-value for the null that the 2006 and 2007 coefficients are jointly zero.

3.1.3 Pooled Difference-in-Differences Estimates

The event studies show how the response evolved over time. To summarize its size in a single number for each group, to compare alternative definitions of childbearing age, and to obtain the estimate we use in Section 4, we pool the years before and after the reform. Firm size is fixed at its 2007 level so that firms' post-reform adjustments cannot move them across the 30-employee threshold. We estimate the following difference-in-differences specification separately for the employment share of each worker group,

$$Y_{jt} = \alpha \, \text{Small}_j + \theta \, (\text{Small}_j \times \text{Post}_t) + \xi_{n(j)t} + \varepsilon_{jt} \tag{3}$$

where Y_{jt} is the share of firm j's workforce in year t that belongs to a given group. We estimate the model separately using five definitions of Y_{jt}, which are the share of all female employees, three narrower definitions restricted to childbearing-age women (ages 18–49, 20–39, and 25–34), and the share of women aged 40 to 59. Small_j equals one if firm j had fewer than 30 employees in 2007, and Post_t equals one for years after 2009. Because the reform took effect partway through 2009, we exclude 2009 from all regressions in this section, so Post_t equals one for 2010–2017 and zero for 2006–2008. $\xi_{n(j)t}$ denotes industry-by-year fixed effects, where $n(j)$ is the industry of firm j. These fixed effects absorb year effects, industry effects, and the main effect of Post_t, so Post_t does not enter separately. ε_{jt} is the error term. The coefficient α captures the time-invariant difference between small and large firms, and the coefficient of interest, θ, captures the change in this difference after the reform. Large firms serve as the control group, and θ identifies the effect of the reform under the assumption that, absent the reform, the share would have evolved in parallel at small and large firms within the same industry. Standard errors are clustered at the firm level.

The results in Table 5 provide baseline evidence that small firms reduced the share of childbearing-age women in their workforce relative to large firms following the 2009 reform. Column (1) shows that the female employment share at small firms fell by 0.48 percentage points relative to large firms after the reform, a modest but statistically significant effect. Correspondingly, the male share rose by 0.48 percentage points. The adjustment is far more pronounced among childbearing-age women. Columns (2) through (4) show that their share fell by 2.69 percentage points under the broad 18–49 definition, 3.49 percentage points under the 20–39 window, and 1.82 percentage points under the 25–34 window, all significant at the 1% level. The coefficient for the 20–39 window is larger than those for both the 18–49 and 25–34 windows, suggesting that the composition response is concentrated among women in their twenties and thirties as a whole and extends beyond the 25–34 core. By contrast, the share of women past childbearing age increased. Column (5) shows that the share of women aged 40 to 59 rose

Table 5: Firms Workforce Composition Responses

	Female Share	Female Childbearing Age Share			Female Non-Childbearing Age Share
	Full	Age 18–49	Age 20–39	Age 25–34	Age 40–59
	(1)	(2)	(3)	(4)	(5)
Small × Post	-0.0048***	-0.0269***	-0.0349***	-0.0182***	0.0240***
	(0.0008)	(0.0008)	(0.0007)	(0.0005)	(0.0006)
Small-firm mean, 2007	0.472	0.339	0.220	0.121	0.190
Industry × year FE	Y	Y	Y	Y	Y
Adj R^2	0.266	0.257	0.218	0.135	0.118
N	4,035,912	4,035,912	4,035,912	4,035,912	4,035,912

Notes: Each column reports the estimate of θ from equation (3). The small-firm mean is the 2007 value from Table 3. These estimates were computed on an earlier version of the analysis sample than the event-study estimates and Appendix Table 11, and will be updated in the next version of the paper. The mean age of women at first birth in Taiwan rose from 28.1 in 2006 to 30.8 in 2017. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

by 2.40 percentage points. Taken together, these results are consistent with small firms shifting their female workforce away from women of childbearing age and toward older women who no longer face fertility-related employment risk, with only a small decline in the overall female share.

3.1.4 Robustness

The decline in the share of childbearing-age women could reflect forces unrelated to the reform. Small firms may have been moving toward older workers in general, for example because their incumbent workforce aged faster or because young workers increasingly sorted into large firms. Such a shift would lower the share of young women at small firms even if the reform did not change how firms treated them. We address this concern by using men of the same age as a comparison group.

Decomposition by age and sex. Table 6 estimates equation (3) separately for women aged 20–39, women aged 40–59, men aged 20–39, and men aged 40–59. Column (5) reports a separate regression in which the outcome is the firm-level difference between the shares of young women and young men, which gives the triple-difference estimate and its standard error. Panel A shows that the share of young men at small firms also fell, by 1.81 percentage points, so some of the decline among young women is part of a broader shift away from younger workers. However, the share of young women fell nearly twice as

much, by 3.49 percentage points, and the triple-difference estimate in column (5), 1.68 percentage points, is precisely estimated. The mirror image appears among older workers. The share of women aged 40–59 rose by 2.40 percentage points, compared with 0.97 percentage points for men of the same age. Net of the change common to both sexes, small firms moved about 1.5 percentage points of their workforce from younger to older women.

Panel B shows that the response specific to women is larger, relative to the total, on the hiring margin. The share of new hires who were young women fell by 1.06 percentage points, compared with 0.37 percentage points for young men, so roughly two thirds of the decline among young women is specific to women (column 5). The hiring share of older men did not change. Because the hiring share reflects whom firms choose to hire in a given year, and not the gradual aging of incumbent workers, this panel provides the most direct evidence of a change in whom small firms hired.

The decline in composition is larger than the decline in hiring alone would produce. A fall of about one percentage point in the hiring share cannot by itself lower the workforce share by 3.5 percentage points within eight years, and the same holds for the female-specific estimates (1.68 versus 0.68 percentage points). The remaining decline must come from other margins, such as more separations of young women from small firms, the aging of incumbent workers, or full-year leave spells that remove women from the annual records. For this reason, we treat the hiring share as our main measure of firms' response and composition as supporting evidence.

3.1.5 Responses by Firm Size

The cost of covering an absence is likely to weigh more heavily on firms with fewer workers who can take over the absent worker's tasks. If so, the response should be strongest at the smallest firms. Our main comparison pools all firms with fewer than 30 employees. Table 7 splits them into three groups, firms with 3–9, 10–19, and 20–29 employees in 2007, and compares each with large firms. The share of young women fell in all three groups by similar amounts, between 3.3 and 3.9 percentage points in the workforce and between 1.0 and 1.6 percentage points among new hires. The main result is not driven by a particular segment of small firms. However, the female-specific component of the composition response declines with firm size, from 1.46 percentage points at firms with 3–9 employees to 0.29 percentage points at firms with 20–29 employees. This pattern is consistent with the cost of an absence weighing more heavily on firms with fewer workers to cover for it, a feature we build into the model in Section 4. On the hiring margin, the female-specific gap is significant in all three groups, ranging from 0.61 to 1.06 percentage points. With finer size groups, Appendix Table 11 shows that the female-specific decline in the hiring share is larger at

Table 6: Workforce Composition and Hiring by Age and Sex

	Women		Men		Difference
	Age 20–39	Age 40–59	Age 20–39	Age 40–59	Women − Men, 20–39
	(1)	(2)	(3)	(4)	(5)
Panel A. Workforce Composition					
Small × Post	-0.0349***	0.0240***	-0.0181***	0.0097***	-0.0168***
	(0.0007)	(0.0006)	(0.0007)	(0.0006)	(0.0011)
N	4,035,912	4,035,912	4,035,912	4,035,912	4,035,912
Panel B. Share of new hires					
Small × Post	-0.0106***	0.0046***	-0.0037***	0.0005	-0.0068***
	(0.0012)	(0.0008)	(0.0012)	(0.0010)	(0.0020)
N	2,413,844	2,413,844	2,413,844	2,413,844	2,413,844
Industry × year FE	Y	Y	Y	Y	Y

Notes: These estimates were computed on an earlier version of the analysis sample than the event-study estimates and Appendix Table 11, and will be updated in the next version of the paper. Each cell reports the estimate of θ, the coefficient on Small × Post, from a separate regression of equation (3). Workers younger than 20 or older than 59 are not assigned to any group, so the coefficients in columns (1)–(4) do not sum to zero. Column (5) is a separate regression whose outcome is the firm-year difference between the shares in columns (1) and (3). Its point estimate equals the difference between those two columns, and its standard error comes from this regression. Panel B is restricted to firm-years with at least one new hire. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

firms with 3 to 14 employees than at firms with 15 to 29 employees.

These results show that the reform lowered the share of young women at small firms by about 1.7 percentage points in the workforce and 0.7 percentage points among new hires, beyond the change experienced by young men, and that this response appears across small firms of all sizes.

Table 7: Responses by Size of Small Firms

	Share of workforce		Share of new hires	
	Women 20–39	Women − Men	Women 20–39	Women − Men
	(1)	(2)	(3)	(4)
3–9 employees × post	-0.0328***	-0.0146***	-0.0101***	-0.0061***
	(0.0007)	(0.0011)	(0.0014)	(0.0023)
10–19 employees × post	-0.0392***	-0.0050***	-0.0153***	-0.0106***
	(0.0008)	(0.0011)	(0.0015)	(0.0025)
20–29 employees × post	-0.0387***	-0.0029**	-0.0162***	-0.0089***
	(0.0009)	(0.0014)	(0.0018)	(0.0031)
Industry × year FE	Y	Y	Y	Y
N	4,035,912	4,035,912	2,413,844	2,413,844

Notes: These estimates were computed on an earlier version of the analysis sample than the event-study estimates and Appendix Table 11, and will be updated in the next version of the paper. Each column reports one regression in which small firms are split by their 2007 employment, with large firms (30 or more employees in 2007) as the omitted group. Columns (2) and (4) are separate regressions whose outcome is the firm-year difference between the shares of women and men aged 20–39. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

3.1.6 Transitions of Female Workers at Childbearing Age

After the reform, small firms employed and hired fewer women of childbearing age. Relative to large firms, the share of these women in small firms' workforce fell by about 3 percentage points, and their share of new hires fell by about 2 percentage points by 2016. These results describe the firms' side of the adjustment. They do not tell us what happened to the women themselves. If small firms became less willing to employ women who might take parental leave, women already working at small firms may have responded by moving to other employers. Large firms are a natural destination. They can more easily absorb the cost of an employee's leave, and they may offer more stable jobs to workers who expect to have children.

We examine this question by tracking where childbearing-age women at small firms go when they

change jobs. Specifically, we ask whether the reform changed the likelihood that these women moved to a large firm, and whether it changed the likelihood that they moved to another small firm. If these women increasingly moved to large firms, the reform would have shifted them toward larger employers. A decline in moves to other small firms would instead suggest that these women found it harder to get hired elsewhere in the small-firm sector. To study these transitions, we restrict the sample to women of childbearing age and estimate the following specification separately for each of the two outcomes,

$$y_{ijt} = \alpha \, \text{Small}_j + \sum_{k \notin \{2008,\, 2009\}} \beta_k \left(\text{Small}_j \times \mathbb{1}[t = k] \right) + \omega_{a(i)t} + \xi_{n(j)t} + \varepsilon_{ijt} \tag{4}$$

where y_{ijt} is an indicator equal to one if woman i, employed at firm j in year t, moves to a different firm in year $t+1$. We consider two outcomes. J2JSmall_{ijt} equals one if she moves to another small firm, and J2JLarge_{ijt} equals one if she moves to a large firm. Small_j indicates whether the origin firm j had fewer than 30 employees in 2007. Women at large firms serve as the control group. The year k runs over 2006, 2007, and 2010–2016, with 2008 omitted; the last year is 2016 because the outcome requires observing the worker in $t+1$.[14] The coefficients of interest, β_k, capture how the likelihood of each transition for women at small firms changed relative to women at large firms in year k, compared with 2008. $\omega_{a(i)t}$ denotes age-band-by-year fixed effects, where $a(i)$ is the age band of worker i, and α and $\xi_{n(j)t}$ are defined as in equation (3). ε_{ijt} is the error term. We cluster standard errors at the level of the origin firm.

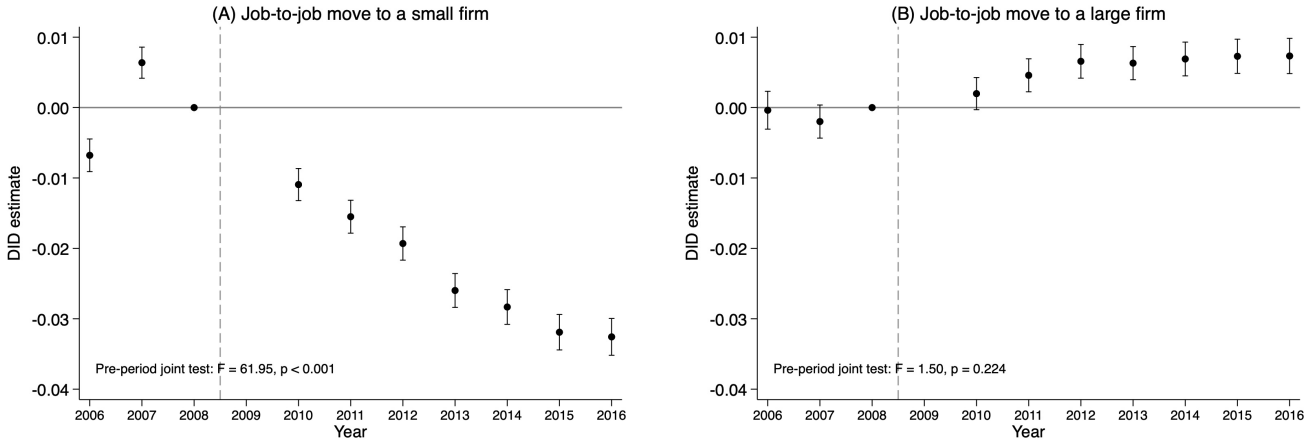
Figure 6 presents the estimated β_k for the two outcomes. Panel (a) shows a large and growing decline in the probability that childbearing-age women at small firms move to another small firm. The coefficients are negative and statistically significant in every post-reform year, falling from about -1.1 percentage points in 2010 to -3.3 percentage points in 2016. Panel (b) shows that the probability of moving to a large firm rose over the same period. The coefficient is small and imprecise in 2010, at about 0.2 percentage points, but becomes positive and significant from 2011 onward, increasing to roughly 0.7 percentage points by 2012 and remaining at that level through 2016.

The pre-reform coefficients are small relative to the post-reform estimates and show no consistent direction. In Panel (a), they are below 0.7 percentage points in absolute value and change sign between 2006 and 2007, while the post-reform effects reach 3.3 percentage points. Given the large sample, these pre-reform coefficients are statistically significant, and we reject the joint null that they are zero ($F = 61.95$, $p < 0.001$). In Panel (b), the pre-reform coefficients are within 0.2 percentage points of zero, and we

[14] Because the outcome for year t is measured in $t+1$, the 2008 reference year captures moves between 2008 and 2009 and is partly exposed to the reform. If the reform had already reduced moves to small firms by 2009, the post-reform coefficients understate the decline.

cannot reject that they are jointly zero ($F = 1.50$, $p = 0.224$).

The two panels suggest that the reform mainly reduced job mobility among childbearing-age women at small firms and redirected only a small part of it toward large firms. This matches the decline in small firms' hiring share of childbearing-age women documented above. If small firms became less willing to hire these women, fewer of them would find a job at another small firm. Some moved to large firms instead. Large firms had already been subject to the parental leave mandate before the reform, so hiring a woman of childbearing age added little to their expected costs. For small firms, the mandate was a new obligation and a relatively more costly one. Job protection also requires one year of tenure with the current employer, so after the reform a woman who planned to have a child had a reason to stay with her employer. This would reduce moves to both small and large firms. Moves to large firms instead rose, so this channel alone cannot explain the pattern.



(a) Destination: small firm

(b) Destination: large firm

Figure 6: Job-to-Job Transitions of Childbearing-Age Women: Destination Small vs. Large Firm

Note: This figure plots the estimated β_k from equation (4) for women aged 20 to 39. The outcome is an indicator for moving from firm j in year t to another small firm (Panel a) or to a large firm (Panel b) in year $t+1$. Small firms are those with fewer than 30 employees in 2007. The reference year is 2008; 2009 is excluded. All regressions include age-band-by-year and industry-by-year fixed effects, and standard errors are clustered at the origin firm. The figure shows 95% confidence intervals.

We examine the sensitivity of these results to violations of parallel trends using the relative magnitudes approach of Rambachan and Roth (2023). This approach allows post-reform deviations from parallel trends to be as large as $\bar{M}$ times the largest change between consecutive pre-reform coefficients. We focus on the average of the post-reform coefficients.[15] For job-to-job moves to small firms, the breakdown

[15]Since 2009 is excluded, 2008 and 2010 are treated as adjacent periods.

value lies between 0.3 and 0.4, meaning the robust confidence set excludes zero as long as $\bar{M}$ is no larger than 0.3. Because the largest pre-reform change, between 2006 and 2007, is sizable, $\bar{M} = 0.3$ already allows for a cumulative bias of about 0.017, roughly 70 percent of the point estimate. The effect remains statistically different from zero even under this bias. Restricting the bias to be negative, which is the direction that would exaggerate the decline, narrows the lower bound of the confidence set but does not change the breakdown value. For moves to large firms, the lower bound reaches zero at $\bar{M}$ of about 0.3, regardless of whether a positive-bias restriction is imposed. Appendix Figure 13 reports the full set of results. Appendix Figure 14 reports event-study estimates for the probability of staying with the same employer and of separating from employment. Separation is difficult to interpret, because it combines several distinct possibilities, including leaving the labor force, taking parental or other leave that appears as a separation in the data, which we cannot distinguish in our data. Changes in separation could arise from very different responses to the reform, and the same ambiguity applies to the probability of staying, which is its complement. For this reason, we do not interpret these estimates causally and report them only for completeness.

3.2 Earnings Growth: Men versus Women of Childbearing Age

The results above show that small firms adjusted how many women of childbearing age they employed and hired, and that these women became less likely to move between jobs. Firms may also respond to a mandate through wages. If employers expect a group of workers to be more costly, they can shift part of that cost onto the workers' pay (Gruber, 1994; Summers, 1989). The decline in job-to-job mobility points in the same direction, since moving to a new employer is an important source of wage growth. We ask whether women of childbearing age at small firms experienced slower wage growth after the reform than comparable men.

We measure wage growth as the year-over-year log difference in wages,

$$\Delta \ln(w_{it}) = \ln(w_{it}) - \ln(w_{i,t-1}),$$

which is observed only for workers with positive earnings in both years.[16] Before 2009, small firms were exempt from the parental leave provisions, while large firms were already covered, so the reform changed the cost of employing women of childbearing age only at small firms. We compare the wage growth of women and men at small and large firms before and after the reform using a triple-difference specification.

[16] We discuss the resulting selection concern below.

We restrict the sample to men and women aged 20–39,

$$\begin{aligned}\Delta \ln(w_{ijt}) = &\sum_{k \notin \{2008,\,2009\}} \beta_k \left(\text{Small}_j \times \text{Female}_i \times \mathbb{1}[t=k]\right) + \sum_{k \notin \{2008,\,2009\}} \pi_k \left(\text{Small}_j \times \mathbb{1}[t=k]\right) \\ &+ \sum_{k \notin \{2008,\,2009\}} \tau_k \left(\text{Female}_i \times \mathbb{1}[t=k]\right) + \psi \left(\text{Small}_j \times \text{Female}_i\right) \\ &+ \chi \,\text{Female}_i + \zeta_j + \omega_{a(i)t} + \xi_{n(j)t} + \varepsilon_{ijt}\end{aligned} \tag{5}$$

where Small_j is defined as in equation (3) and Female_i indicates that worker i is a woman. The terms π_k and τ_k absorb year-specific changes in wage growth common to all workers at small firms and to all women, and χ and ψ absorb time-invariant gender differences in wage growth and how they differ at small firms. Firm fixed effects ζ_j absorb the main effect of Small_j, and $\omega_{a(i)t}$ and $\xi_{n(j)t}$ are the age-band-by-year and industry-by-year fixed effects defined in equations (4) and (3). The year k runs over 2007 and 2010–2017; 2007 is the first year because wage growth requires earnings in $t-1$. The coefficients of interest, β_k, capture how the gap in wage growth between women and men at small firms changed in year k, relative to the same gap at large firms and relative to 2008. Standard errors are clustered at the firm level.

Appendix Figure 15 reports the estimated β_k. Among all workers employed in both years, the post-reform estimates are small and mostly negative, between -0.7 and 0.1 log points, and none is statistically significant at the 5 percent level. However, the 2007 estimate is positive and significant, at 0.9 log points, so the slightly negative post-reform estimates may partly continue a trend that began before the reform.

Because the reform reduced job-to-job mobility among childbearing-age women at small firms (Section 3.1.6), lower mobility alone could reduce their average wage growth even if their pay at a given employer did not change. We repeat the estimation for workers who stay with the same employer between $t-1$ and t. The estimates for stayers are less precise, ranging from -1.7 to 2.0 log points, and none of them is statistically significant. Because wage growth is observed only for workers employed in both years, these estimates cannot rule out selection arising from workers who leave employment altogether.

We find no clear evidence that the reform slowed the wage growth of childbearing-age women at small firms relative to men. Combined with the hiring results, this suggests that small firms dealt with the cost of the mandate by hiring fewer women of childbearing age. They do not appear to have cut the pay of the women already on their payroll.

3.3 Workers Labor Market Outcomes

The previous sections show that, after the reform, women of childbearing age made up a smaller share of the employees and new hires of small firms. We now turn to the women who were already working at small firms when they had a child. The reform gave these mothers the right to job-protected leave, which could help them keep their jobs after birth. However, if their employers came to view them as more costly, the reform could also have weakened their position at these firms. To see which effect dominates, we compare mothers whose pre-birth employer was a small firm with mothers at large firms, before and after the reform.

We measure each mother's outcomes at a fixed horizon s, defined as the number of years since she gave birth, so that $s = 0$ is the year of birth, $s = 1$ the following year, and so on. Each horizon is a separate outcome, so each regression uses one observation per mother.

For each horizon $s \in \{0, 1, \ldots, 7\}$, we estimate

$$y_{i,s} = \theta_s \left(\text{Small}_i \times \text{Post}_i\right) + \alpha_s \, \text{Small}_i + \mu_{t,s} + \sigma_{a,s} + \varepsilon_{i,s}, \tag{6}$$

where i indexes mothers and s is the number of years since birth. The outcome $y_{i,s}$ is either an indicator for being employed or annual earnings in year $b_i + s$, where b_i is mother i's birth year. Earnings are annual wage income reported in tax records, measured in thousands of USD, including zeros for years with no wage income and excluding the parental leave allowance. Under this definition, a mother on leave for an entire year has zero earnings and is counted as not employed, and a mother on leave for part of the year has lower earnings. Small_i equals one if the mother's employer in year $b_i - 1$ had fewer than 30 employees in 2007, and Post_i equals one if $b_i \geq 2010$; births in 2009 are excluded. We index Small_i by mother because each mother is assigned to a firm through her pre-birth employer. $\mu_{t,s}$ are calendar-year fixed effects and $\sigma_{a,s}$ are fixed effects for the mother's age. Because s is fixed within each regression, calendar year equals $b_i + s$, so the year fixed effects are equivalent to birth-cohort fixed effects and absorb the main effect of Post_i. Standard errors are clustered at the level of the pre-birth employer.

The coefficient θ_s is the difference-in-differences estimate at horizon s, the change in the gap between mothers at small and large firms for births after the reform, relative to births before it. Identification rests on the parallel trends assumption that, absent the reform, this gap would have evolved the same way for pre- and post-reform cohorts.

Equation (6) compares all pre-reform cohorts with all post-reform cohorts, so it cannot show whether the gap between mothers at small and large firms was already changing before the reform. To examine pre-reform trends, we hold the horizon fixed at $s = 0$ and replace $\text{Small}_i \times \text{Post}_i$ in equation (6) with

interactions between Small_i and indicators for each birth cohort,

$$y_{i,s} = \sum_{c \neq 2008} \beta_{c,s} \left(\text{Small}_i \times \mathbb{1}[b_i = c] \right) + \alpha_s \, \text{Small}_i + \mu_{t,s} + \sigma_{a,s} + \varepsilon_{i,s}, \tag{7}$$

where all terms are defined as in equation (6). We omit the 2008 cohort, the last cohort to give birth before the reform, so each $\beta_{c,s}$ measures the gap between mothers at small and large firms in cohort c relative to the same gap in 2008. If the parallel trends assumption holds, the coefficients for pre-reform cohorts should be close to zero. Because treatment is defined by the employer in the year before birth and our data begin in 2006, 2007 is the only pre-reform cohort besides the omitted one, so the test of pre-reform trends reduces to testing $\beta_{2007,s} = 0$. We focus on the year of birth ($s = 0$) because it is the only horizon at which the outcome years of both pre-reform cohorts, 2007 and 2008, fall entirely before the reform. At $s \geq 1$, the 2008 reference cohort is observed in 2009 or later and is partially treated, because the reform also covered parents of children under age three at the time it took effect. Figure 7 plots the estimates for employment and earnings and reports the F-statistic and p-value for this test. We cannot reject the null for either outcome ($p = 0.67$ for employment and $p = 0.11$ for earnings). While this test does not reject the parallel trends assumption, it is based on only a single pre-reform cohort, so its power to reject is less than ideal.

The cohort estimates also show how the gap at $s = 0$ evolved after the reform. For employment, the post-reform coefficients are small and mostly insignificant; only the 2013 coefficient, at -1.3 percentage points, is significant at the 5 percent level. For earnings, the 2010 and 2011 cohorts show small and insignificant differences. From the 2012 cohort onward, mothers at small firms earn about 0.8 to 1.2 thousand USD less in the year of birth than mothers at large firms, relative to the 2008 cohort, and all of these coefficients are statistically significant.

Figure 8 reports $\hat{\theta}_s$ for each horizon. The reform has no detectable effect on employment in the year of birth. One year after birth, employment of mothers at small firms falls by 1.8 percentage points relative to mothers at large firms. The effect shrinks to 1.3 percentage points at $s = 2$ and 0.8 percentage points at $s = 3$, and is close to zero and statistically insignificant from $s = 4$ onward. Panel B reports the estimates for earnings. Unlike employment, earnings of mothers at small firms fall already in the year of birth, by 0.67 thousand USD relative to mothers at large firms. The decline peaks at 0.84 thousand USD at $s = 2$, shrinks to 0.45 thousand USD at $s = 4$, and is smaller and less precisely estimated from $s = 5$ onward. Because employment does not change at $s = 0$, the decline in the year of birth is driven by lower earnings among mothers who remain employed. This pattern is consistent with mothers at small firms taking longer leave after the reform. Earnings exclude the leave allowance, so months spent on

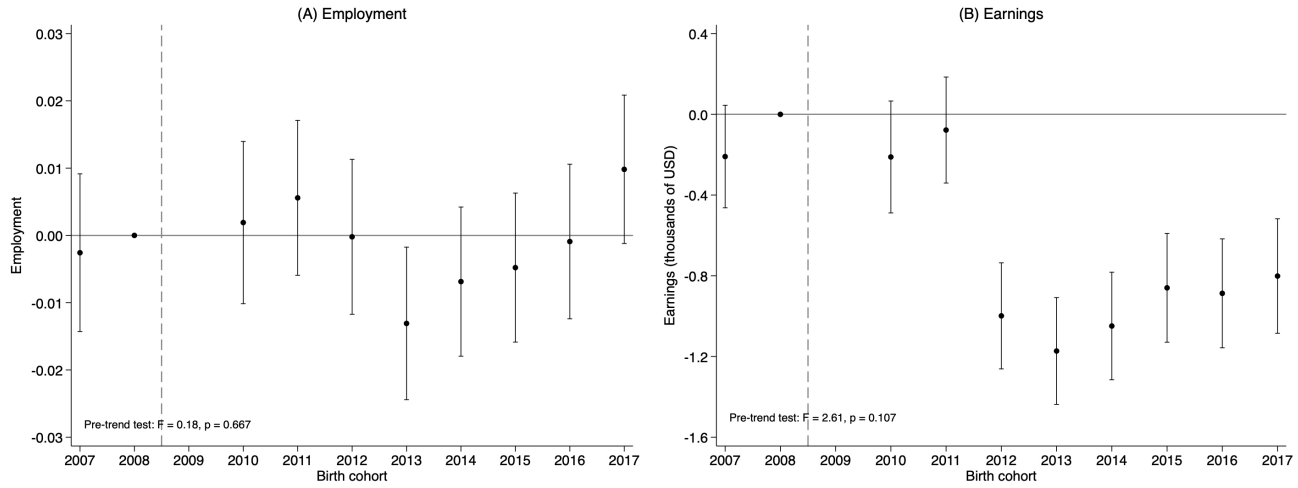


Figure 7: Pre-Reform Trends by Birth Cohort, Mothers

Notes: The figure plots estimates of $\beta_{c,s}$ from equation (7) at horizon $s = 0$ with 95% confidence intervals. Each coefficient is the employment (Panel A) or earnings (Panel B) gap between mothers at small and large firms in birth cohort c, relative to the same gap in the 2008 cohort. The dashed line marks the 2009 reform. The pre-trend test reports the F-statistic and p-value for $H_0 : \beta_{2007,s} = 0$. All regressions include calendar-year and age fixed effects. Standard errors are clustered at the level of the pre-birth employer.

leave lower annual earnings without changing measured employment. However, the cohort estimates call for caution. The decline at $s = 0$ appears only from the 2012 cohort onward, and the 2007 coefficient, although not statistically significant, is similar in size to those for the 2010 and 2011 cohorts. At longer horizons, mothers in the 2007 and 2008 cohorts become partially treated once their outcome year passes 2009, so θ_s is biased toward zero.

The analysis for fathers shows that the reform has no detectable effect on their earnings in the year of birth (Appendix Figure 16). The earnings coefficients fluctuate around zero, between -0.34 and 0.31 thousand USD, and only the 2013 estimate is marginally significant at the 5 percent level. For employment, the post-reform coefficients are positive and statistically significant but small, about 1 percentage point from 2010 to 2015. The 2007 coefficient is of similar size, 0.7 percentage points, and is also significant, so the employment gap between fathers at small and large firms was already moving before the reform. The estimates rise to 1.9 and 3.2 percentage points in 2016 and 2017. Appendix Figure 17 reports the estimates by years since birth. In contrast to mothers, we find no evidence that the reform lowered the employment or earnings of fathers at small firms.

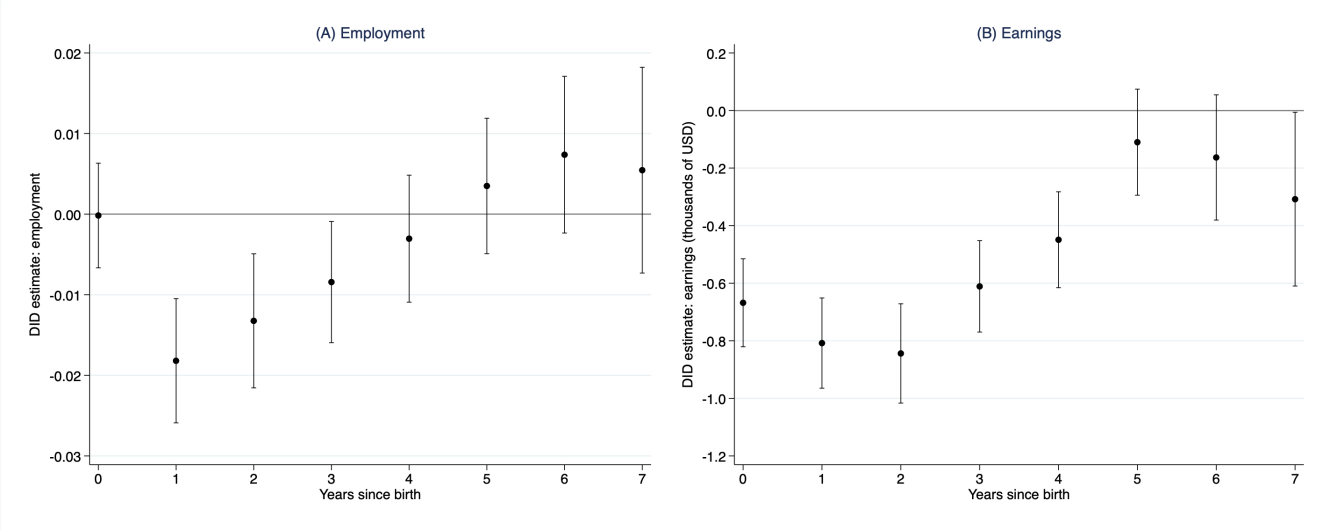


Figure 8: Difference-in-Differences Estimates by Years Since Birth, Mothers

Notes: Each point is $\hat{\theta}_s$ from a separate estimation of equation (6) at horizon s, with 95% confidence intervals. Panel A is employment; Panel B uses annual earnings in thousands of USD. All regressions include calendar-year and age fixed effects. Standard errors are clustered at the level of the pre-birth employer.

3.4 The Cost of Holding a Slot

The results in this section show that small firms responded to the reform mainly by hiring fewer women of childbearing age, and their relative pay changed little. To interpret this response, we first need to be precise about what the mandate required. Firms had to keep a job open while an employee was on parental leave. During that time, the firm cannot fill the position permanently and must either leave the work undone or cover it at extra expense. We refer to this as the cost of holding a slot. Pre-reform survey evidence gives a sense of what this cost involves. The Council of Labor Affairs' 2007 Survey on Female Employment Management, which covered establishments with 30 or more employees, asked employers how they managed when a worker went on parental leave. Most of them (61.7 percent) did not hire a replacement and instead redistributed the work among colleagues in the same department.[17] As a result, the cost of a held slot falls mostly on coworkers, who pick up the extra work through overtime or leave some of it undone. Few firms pay for it by hiring a temporary replacement. These establishments were large and already subject to the mandate. A firm with fewer than 30 employees has fewer workers over whom to spread an absent employee's duties, so the same leave is likely to weigh more heavily on it, which may explain why the hiring response is concentrated among small firms. Because small firms adjusted

[17] Council of Labor Affairs (2008), "Implementation of Parental Leave without Pay," *Labor Statistics Bulletin*, July 4, 2008, based on the 2007 Survey on Female Employment Management.

mainly through hiring, the size of the drop in their hiring of childbearing-age women indicates how costly firms expected these women to be. In the next section, we use a simple search model in which firms bear a cost for each woman on leave, and we ask whether the cost of holding a slot alone can account for the observed hiring response.

4 Model

In this section, we extend the Burdett–Mortensen model to incorporate gender equal pay and job-protected leave, and we use it to interpret the hiring response of small firms and to quantify the cost of leave required to account for it. Firms post wages in a labor market with on-the-job search. Childbearing-age women differ from other workers only in that they may give birth. A woman who gives birth while employed takes job-protected leave, and her employer pays a flow cost to keep her position open until she returns. Taiwan's Gender Equality in Employment Act requires employers to pay women and men equally for equal work, so a firm pays childbearing-age women the same wage as other workers in the same job. A firm cannot pass the expected cost of leave on through lower pay, and its only margin of adjustment is whether to hire these women at all. We also allow the cost of covering an absence to fall with the number of coworkers who can take over the absent worker's duties, so that the same leave weighs more heavily on a firm with few employees. A firm hires a childbearing-age woman only if the surplus it earns on her covers her expected leave cost, and the firms that stop hiring her are those with low surplus and few coworkers, which are also the small firms. The model links the share of firms that stop hiring childbearing-age women to the distribution of firm surplus, which we recover from the wage distribution, and so lets us back out the cost of leave that reproduces the hiring gap.

4.1 Environment

Time is continuous. There are two permanent worker types indexed by $g \in \{n, y\}$. Type n pools men and non-childbearing-age women into a single type, and type y is childbearing-age women, with masses $N_n + N_y = 1$. Types are permanent. In reality, a woman hired at a young age eventually ages out of her childbearing years, so the expected leave cost of a new hire falls over the course of the match. Treating the type as permanent keeps the model simple. For a given cost per day of leave, it overstates the expected leave cost of a young hire, at least somewhat. This does not affect the wedge we back out below, which depends only on firms' rents and the share of firms that do not hire women of childbearing age. It does mean that our benchmark for the direct cost of an absence, which uses the current share of women on

leave, somewhat overstates the expected direct cost of hiring a young woman. Workers are risk neutral, ex ante identical within type, discount at rate $r > 0$, cannot save or borrow, and search both off and on the job at no cost. Workers not exposed to the leave policy are treated symmetrically, and the two types differ only in their exposure to fertility, their separation rates δ_g, and their flow values of non-employment b_g.

Instantaneous utility is $u(x, n) = x - \varsigma n$, where x is consumption, $n \in \{0, 1\}$ indicates whether the worker is at work, and $\varsigma > 0$ is the disutility of work. Consumption is

$$x = \begin{cases} w & \text{if employed and at work,} \\ \nu & \text{if on parental leave,} \\ b_g & \text{if unemployed,} \end{cases}$$

where w is the wage, ν is the government-funded leave benefit, and b_g is the flow value of non-employment for type $g \in \{n, y\}$.

Search. All workers search in a single, unsegmented market. An unemployed worker meets firms at the exogenous rate λ_0 and an employed worker at the exogenous rate λ_1, and both types share these contact rates. Jobs are destroyed exogenously at the type-specific rate δ_g. We allow separation rates to differ because we measure them directly for each type. Contact rates are common by assumption, so that differences in job finding between the two types reflect firms' hiring decisions. If women of childbearing age had lower contact rates, for example because they search less, the model would attribute the resulting gap in job finding to the hiring rule, and the implied cost of leave would be overstated.

Fertility and leave. Type-y women have a child at Poisson rate $\kappa > 0$; type n has $\kappa = 0$. An employed woman who gives birth takes job-protected paid leave. While on leave, she stops working and searching and receives the government-funded benefit ν. She returns to her pre-leave job at the same wage at Poisson rate γ, so a leave spell lasts $1/\gamma$ on average. Every woman on leave eventually returns, and γ governs how long the leave lasts, not the share of women who return. Job-protected leave applies only to employed women, so a birth while unemployed does not change a woman's search status. The steady-state share of attached type-y women who are on leave at a point in time is

$$\phi = \frac{\kappa}{\gamma + \kappa} \in [0, 1).$$

Firm-borne leave cost. The benefit ν is government funded, but a firm pays a flow cost for each woman on leave, including the holding-slot cost and the cost of covering her work through coworkers'

overtime or a temporary hire. This cost depends on how many coworkers can share the absent worker's duties. A firm with m coworkers pays

$$c(m) = \frac{\bar{c}}{(1+m)^{\eta}}, \qquad \eta \geq 0, \tag{8}$$

so that $\bar{c}$ is the cost borne by a firm with no one to cover the absence, and η governs how fast the cost falls with the number of coworkers. The case $\eta = 0$ is a single cost common to all firms. Throughout this section we treat the post-reform universal regime, in which every firm bears this cost.

4.2 Firms

A firm has productivity $p \sim \Gamma$ on $[\underline{p}, \bar{p}]$ and posts a wage $w(p)$. Under the equal-pay provision, a firm pays a single wage $w(p)$ to both types of workers. We assume that this wage is set by the firm's type-n workforce, so $w(\cdot)$ is the standard single-type Burdett–Mortensen schedule (Proposition 4 in the Appendix). Childbearing-age women make up about a fifth to a quarter of employees and new hires (Table 3), so we treat this as an approximation. We impose it because it gives the wage schedule a closed form, which we need to recover rents from the data. The assumption rules out a firm lowering the wage it pays all workers to offset the expected cost of leave, and we adopt it to keep the model tractable. Our finding that the wage growth of women of childbearing age did not change relative to men is consistent with it but does not test it, since a wage cut for all workers at small firms would not appear in that comparison. As in Burdett and Mortensen (1998), more productive firms post higher wages and attract and retain more workers, so firm size increases with p.

The hiring margin. A firm posts its wage before meeting any worker. When it meets a worker, it observes her type and decides whether to make an offer at the posted wage. Because the wage is already fixed, the decision to hire a type-y woman depends only on whether the flow surplus she generates covers her expected flow leave cost. For each attached type-y woman, a firm earns $(1-\phi)\big(p-w(p)\big)$ in production surplus and pays $\phi\, c(m)$ in leave cost. Since the wage is set by the type-n margin, the firm cannot offset this cost by paying her less, and it hires type-y workers only if its margin on them is non-negative,

$$(1-\phi)\big(p-w(p)\big) - \phi\, c(m) \;\geq\; 0 \qquad \Longleftrightarrow \qquad \rho(p) \;\geq\; \Delta(m) \equiv \frac{\phi\, c(m)}{1-\phi} = \frac{\bar{\Delta}}{(1+m)^{\eta}}, \tag{9}$$

where $\rho(p) \equiv p - w(p)$ is the firm's rent per worker and $\bar{\Delta} \equiv \phi\bar{c}/(1-\phi)$. The wedge $\Delta(m)$ is the expected leave cost per childbearing-age woman per year of employment, scaled by the share of time she is at work. Type-n workers do not take leave, so every firm hires them.

Proposition 1 (Hiring rule)**.** *Define a firm's breakeven cost* $z \equiv (1+m)^{\eta}\rho(p)$*. A firm hires childbearing-age women if and only if* $z \geq \bar{\Delta}$*. If* z *is increasing in* p*, there is a unique threshold* p^{**} *with* $z(p^{**}) = \bar{\Delta}$*, and firms with* $p < p^{**}$ *hire no childbearing-age women. The wage* $w(p)$ *does not depend on* $\bar{c}$*,* η*, or* ϕ*.*

A firm stops hiring childbearing-age women when its rent is low and it has few coworkers to cover an absence. In the model, firm size rises with productivity, so these firms are small. In the data, firms that pay similar wages vary in size, so we assign each firm the rent implied by its wage in (12) below and its observed employment. Because the wage is set by the type-n margin and equal pay prevents firms from paying childbearing-age women less, the reform should change whom small firms hire but leave their wages unchanged. In Section 3, we find no clear change in the wage growth of childbearing-age women at small firms relative to men. Our tax data report annual earnings but not hours, so we cannot examine hourly wages directly.

The model also implies that, once leave is mandated everywhere, the firms that do not hire women of childbearing age are small. Our reduced-form design does not test this cross-sectional prediction. It compares small firms with large firms that were already covered, and we do not observe how large firms responded when the mandate first applied to them in 2002. What the design does test is the model's prediction that extending the obligation to small firms lowers their hiring of women of childbearing age. The finer comparison within small firms in Section 3.1.5, where the response is larger at the smallest firms, is consistent with a cost that falls with the number of coworkers.

4.3 Offer distributions

Type n receives an offer from every firm it meets. A firm that does not hire childbearing-age women makes no offer when it meets a type-y woman, so for her such a meeting is equivalent to not meeting a firm at all. Let $H(\bar{\Delta}) \equiv \Pr(z < \bar{\Delta})$ denote the share of firms, weighted by the contact distribution Γ, that do not hire type-y workers. By the thinning property of Poisson processes, contacts that lead to an offer arrive at rate

$$\tilde{\lambda}_{jy} = \lambda_j\big[1 - H(\bar{\Delta})\big], \qquad j \in \{0, 1\}. \tag{10}$$

A tilde denotes an equilibrium object. Although λ_0 and λ_1 are common to both types, childbearing-age women receive fewer offers than type-n workers, both on and off the job. Appendix D gives the full derivation and the distribution of the offers they do receive.

It follows that childbearing-age women find jobs more slowly than other workers. Let JF_g denote the job-finding rate of type g. Every offer pays the posted wage $w(p)$, which is at least the minimum wage

and, we assume, at least the reservation wage of either type, so every offer is accepted and

$$\frac{\mathrm{JF}_y}{\mathrm{JF}_n} = \frac{\tilde{\lambda}_{0y}}{\lambda_0} = 1 - H(\bar{\Delta}). \tag{11}$$

Given an observed relative job-finding rate R, the cost scale $\bar{\Delta}$ is the $(1-R)$ quantile of the distribution of breakeven costs z across firms.

4.4 Taking the Model to the Data

The calculation does not require solving for the full equilibrium. It uses three restrictions of the model: the firm's first-order condition for the wage, which combined with the steady-state relation between the earnings and offer distributions gives each firm's rent as a function of observed wages; the hiring rule in Proposition 1; and the relation in (11) between relative job-finding rates and the share of firms that do not hire women of childbearing age. It also assumes that every offer is accepted. We do not need to solve for the wage schedule, the productivity distribution, reservation wages, or equilibrium firm sizes, because we observe each firm's wage and number of coworkers directly. Appendix D defines the steady-state equilibrium and characterizes these remaining objects, showing that the three restrictions come from a complete equilibrium.

We use equation (11) to back out the cost of leave that reproduces the observed hiring gap, and we compare this cost with the direct cost of covering an absence. Doing so requires the relative job-finding rate, each firm's rent ρ, the number of coworkers m, and a value of η.

The relative job-finding rate. We use two values of $\mathrm{JF}_y/\mathrm{JF}_n$. The first is the raw ratio after the reform, $\hat{R} = 0.579$ (s.e. 0.005), which we compute from the annual transition matrices of childbearing-age women and type-n workers. We apply it to all firms. Since the model attributes the whole gap to the cost of leave, this ratio gives an upper bound on that cost. Part of the raw gap reflects labor supply, because many non-employed women of childbearing age, in particular new mothers, are not looking for work. The second ratio keeps only the part of the gap that the reform caused and that is specific to women. It comes from the difference-in-differences estimate for the share of women aged 20 to 39 among new hires at small firms, net of the same share for men of that age (Table 6, Panel B, column 5). Subtracting the share of young men removes shocks that hit young workers of both sexes at small firms after the reform. The pre-reform share is $s_0 = 0.251$ and the female-specific decline is 0.0068, so the odds that a new hire is a childbearing-age woman fall to $\hat{R}^{\mathrm{DiD}} = 0.964$ (s.e. 0.010) of their pre-reform level. This estimate comes from an earlier version of the analysis sample. The event-study estimates in Figure 3, based on the final

sample, suggest a somewhat larger decline, which would raise the implied wedge. The reform raised the cost of leave only at small firms, so we apply this ratio to the distribution of z among small firms.

Firm rents. We recover rents from the wage distribution of type-n workers employed by the same employer for a full calendar year, following Bontemps et al. (2000). Because $w(\cdot)$ is the single-type Burdett–Mortensen schedule, the firm's first-order condition, combined with the steady-state relation $G = F/(1+k\bar{F})$ between the wage distribution of employed workers G and the offer distribution F, gives the rent of the firm posting wage w as a function of observables,

$$\rho(w) = \frac{1 + k\,G(w)}{2k\,g(w)}, \qquad k \equiv \frac{\lambda_1}{\delta_n}. \tag{12}$$

We measure each firm's wage as the mean annual earnings of its full-year type-n workers and assign it the rent at that wage. We estimate the density g from a histogram with NT\$5,000 (about US\$157) bins, smoothed over three adjacent bins. Kernel estimates are biased downward near the lower truncation point of the wage distribution and would overstate rents there by up to 80 percent (Appendix D.9).

The arrival-rate ratio k. The ratio k governs the size of the rent in (12). Annual transition rates cannot recover it (Appendix D.8), so we estimate it from two wage distributions instead. Workers hired out of non-employment sample the offer distribution F, while full-year stayers follow G. Rearranging the steady-state relation gives

$$F(w) - G(w) = k\,G(w)\big[1 - F(w)\big], \tag{13}$$

which we estimate by least squares without a constant across NT\$5,000 wage bins above the minimum-wage region. We measure F from the first full calendar year of earnings of type-n workers hired out of non-employment who stay with the new employer for at least two years. The estimate is $\hat{k} = 0.44$ (s.e. 0.010), within the range of 0.3 to 2 that Jolivet et al. (2006) report for eleven countries. The ratio $k = \lambda_1/\delta_n$ is the expected number of outside offers a worker receives before her job ends, and $1/(1+k)$ is the probability that she receives none. With $k = 0.44$, about 70 percent of workers never receive a competing offer during a job, so firms face little competition for their employees and earn large rents. With $k = 2$, only a third of workers never receive one, competition is stronger, and rents are smaller. The estimate is not sensitive to the lower cutoff of the wage distribution. Requiring hires to stay two years selects those with higher wages, which moves F toward G and biases $\hat{k}$ downward. A lower k raises the implied rents, so this bias works in the same direction as the upper-bound interpretation below. We also report results for $k = 1$ and $k = 2$, two values within the range estimated in the literature.

The minimum wage. The minimum wage binds at the bottom of the wage distribution, and the first-order condition behind (12) does not hold where it binds. We exclude the spike of earnings at the minimum wage and use (12) only above $w_b = 1.2$ times the annual minimum wage, the point at which the spike ends. Eleven percent of full-year type-n workers earn less than w_b. Firms whose mean wage lies below w_b, 15.9 percent of all firms and 18.2 percent of small firms, are assigned the rent at w_b. Their true rent could be anywhere between zero and this value, so the assigned rent is an upper bound.

Coworkers and η. We measure m as a firm's employment in 2007 minus one and report results for $\eta \in \{0, 0.5, 1\}$. The data favor $\eta > 0$. Splitting small firms into six size groups, the female-specific decline in the share of new hires averages 0.87 percentage points at firms with 3 to 14 employees and 0.54 percentage points at firms with 15 to 29 employees, and the difference is statistically significant ($p = 0.001$; Appendix Table 11). This fits the 2007 survey evidence that firms cover most absences by spreading the work among coworkers.

The direct cost benchmark. To assess the magnitude of Δ, we compare it with the wedge implied by the direct cost of an absence. Suppose that covering one day of leave costs the firm about one day of wages. We consider two values, NT$1,000 (about US$31 at the 2015 exchange rate), which is close to the daily minimum wage and equals the payment the government has made since January 2026 to small employers for each day of parental leave they grant, and NT$2,000 (about US$63), which is close to the average daily wage. With 250 working days per year, these costs give an annual flow cost c, and by (9) the implied wedge is $\Delta^{\text{direct}} = c\,\phi/(1-\phi)$. We compute ϕ from the observed rate at which female employees begin parental leave. In Figure 1, an average of 2.2 percent of female employees aged 25 to 44 began leave each year in 2010–2016, a rate that already accounts for both fertility and take-up. Combined with the average leave length of 7.1 months reported by the government,[18] this gives $\phi/(1-\phi) = (0.022/12) \times 7.1$ and $\phi = 0.0129$. Because $\hat{\Delta}$ does not depend on ϕ, the choice of leave rate and leave length affects only the benchmark.

4.5 How Large Would Leave Costs Have to Be?

Table 8 reports the results. The first row of Panel A reports the cost scale $\bar{\Delta}$, the wedge faced by a firm with no coworkers. The second reports the mean wedge across the firms to which each ratio applies,

[18] Directorate-General of Budget, Accounting and Statistics, Executive Yuan, "Overview of Parental Leave" (in Chinese), 2016, https://ws.dgbas.gov.tw/public/data/dgbas03/bs2/gender/analysis/employment%20economy%20and%20welfare/%E8%82%B2%E5%AC%B0%E7%95%99%E8%81%B7%E5%81%9C%E8%96%AA%E6%A6%82%E6%B3%81.pdf.

all firms for the raw gap and small firms for the reform-induced gap, which is the relevant object for comparison with Panel B.

Table 8: Implied expected leave cost per childbearing-age woman

	Raw gap, $\hat{R} = 0.579$, all firms			Reform-induced, $\hat{R}^{\text{DiD}} = 0.964$, small firms		
	$\eta = 0$	$\eta = 0.5$	$\eta = 1$	$\eta = 0$	$\eta = 0.5$	$\eta = 1$
Panel A. Implied by the hiring gap						
Cost scale $\hat{\bar{\Delta}}$ (US$)	19,502	60,360	179,729	15,722	28,300	49,018
	(22)	(573)	(2,702)	(45)	(977)	(2,442)
Mean $\hat{\Delta}(m)$ (US$)	19,502	19,918	23,138	15,722	10,516	7,445
	(22)	(189)	(348)	(45)	(363)	(371)
Mean $\hat{\Delta}(m)$ / min. wage	2.70	2.76	3.20	2.18	1.46	1.03
Panel B. Implied by the direct cost of absence, Δ^{direct} (US$)						
NT$1,000 per leave day: 103		NT$2,000 per leave day: 205				

Notes: All values are annual, in 2015 US$, converted from 2015 NT$ at the 2015 annual average exchange rate of 31.898. The annual minimum wage is US$7,226 (NT$230,505). $\hat{\bar{\Delta}}$ is the $(1-\hat{R})$ quantile of the breakeven cost $z = (1+m)^{\eta}\rho$ across firm-years, where ρ is the rent in (12) and m is 2007 employment minus one. The mean wedge is $\hat{\bar{\Delta}}$ times the mean of $(1+m)^{-\eta}$ across all firms (raw gap) or small firms (reform-induced). $\hat{k} = 0.44$ and $w_b = 1.2$ times the annual minimum wage. In the reform-induced columns, the quantile falls among firms paying near the minimum wage, whose rent is set to its upper bound, so these figures are upper bounds. Standard errors in parentheses reflect sampling variation in $\hat{R}$ only. The sample has 1,069,082 firm-years, 883,452 of them at firms with fewer than 30 employees in 2007. Panel B uses $\Delta^{\text{direct}} = c\,\phi/(1-\phi)$ with 250 working days per year, the rate at which female employees begin parental leave in Figure 1, and a mean leave length of 7.1 months. Results for other values of k and w_b are reported in Appendix Table 12.

A single cost common to all firms ($\eta = 0$) requires a large wedge. To reproduce the raw gap, the marginal firm would have to expect leave-related costs of 2.7 times the annual minimum wage for each year it employs a childbearing-age woman, and the reform-induced gap requires 2.2 times. These amounts far exceed the wage cost of covering the days of leave, which is at most 2.8 percent of the annual minimum wage (Panel B). The wage cost is a narrow benchmark, however. Leave can also force a firm to put projects on hold, recruit and train a temporary replacement, or reorganize work among the remaining staff, and these disruptions are harder to measure.

Allowing the cost to fall with the number of coworkers separates the two gaps. For the reform-induced gap, the mean wedge among small firms falls from 2.2 times the minimum wage at $\eta = 0$ to 1.0 times at

$\eta = 1$. At $\eta = 1$, this is about US$7,400 (NT$237,000) per year of employment.[19] This is large relative to the wage cost of covering an absence, but it is within the range of what the broader disruptions described above could cost a small firm. The reform made small firms bear the cost of leave, and the firms that responded are those with the fewest coworkers to cover an absence. For the raw gap, the mean wedge rises with η, from 2.7 to 3.2 times the minimum wage. The raw gap is present at firms of every size, including large firms with many coworkers who could cover an absence, and a cost that falls with the number of coworkers cannot generate it without a very large cost scale. Difficulty covering an absence can account for part of the response to the reform, but not for the gap in hiring that existed regardless of the reform. That gap may reflect differences in labor supply, since many non-employed mothers are not looking for work, as well as costs that firms attach to employing women of childbearing age beyond the leave itself, such as exits after childbirth, reduced hours or effort on return, and the prospect of further births. The model captures none of these. Sun (2026) finds a similar pattern in China, where wage responses to an increase in expected fertility are concentrated in sectors with persistent employment relationships, which suggests that employers price expected future productivity more than replacement costs.

The estimates for the reform-induced gap are upper bounds. The 3.6 percent of small firms that stop hiring childbearing-age women all pay close to the minimum wage, where the first-order condition does not hold and the rent could be close to zero. As a result, the data cannot rule out that a cost as small as the direct cost of an absence accounts for the reform-induced response, provided that it falls on firms with low rents and few coworkers. The model's prediction that the response is concentrated among such firms is supported by Appendix Table 11. All estimates in Panel A of Table 8 also overstate the cost that firms attach to leave because we recover rents from the full cross-sectional distribution of type-n wages. As Hornstein et al. (2011) show, observed worker flows place tight limits on how much wage dispersion search frictions can generate, and much of the dispersion in raw wages comes from worker heterogeneity. Treating all of it as frictional overstates the rent that firms earn. Recovering rents from wages adjusted for worker heterogeneity, for example from the firm effects in a two-way fixed effects decomposition of wages, would tighten these bounds, and we leave this extension for future work.

[19] The minimum wage binds for a nontrivial share of workers in Taiwan. Eleven percent of full-year type-n workers earn less than 1.2 times the annual minimum wage (Appendix D.9). Converted at the market exchange rate, Taiwanese earnings also appear low relative to those in the United States. In terms of purchasing power, they are considerably higher.

5 Conclusion

Parental leave mandates protect the jobs of new parents, but the firms that grant the leave bear part of its cost. We study how workers and firms responded to Taiwan's 2009 reform, which extended job-protected leave to firms with fewer than 30 employees while financing paid leave outside the firm. Large firms were already required to hold jobs open, so comparing small and large firms before and after the reform identifies the effect of extending this obligation to small firms.

After the reform, small firms hired fewer women of childbearing age. By 2016, the share of new hires who are women aged 20 to 39 had fallen by about 2 percentage points relative to large firms. The share of young men fell by much less, so most of this decline is specific to women. Over time, the lower hiring shifted the composition of their workforce. The share of women aged 20 to 39 fell by 3.5 percentage points, while the share of older women rose.

The reform also lowered the employment of mothers at small firms after childbirth. One year after the first birth, their employment fell by 1.8 percentage points relative to mothers at large firms, and the gap faded within a few years. Their earnings fell already in the year of birth, which may reflect longer leave. We find no evidence that the reform lowered the employment or earnings of fathers.

Women of childbearing age at small firms also lost job mobility. They became less likely to move to another small firm and only slightly more likely to move to a large firm, which had been covered all along. We find no clear evidence that their wage growth slowed relative to men.

To interpret the firm response, we extend the search model of Burdett and Mortensen (1998) to include job-protected leave and gender equal pay, which prevents firms from paying women of childbearing age less than other workers. As a result, a firm hires them only when the rent it earns on a worker exceeds her expected leave cost. In the model, the difficulty of covering an absence at firms with few employees can explain the decline in hiring after the reform. It cannot explain the gap in job finding that exists at firms of all sizes, which may reflect differences in labor supply as well as other costs that firms expect from employing these women, such as exits after childbirth, shorter hours on return, or additional births.

Since January 2026, Taiwan has paid employers with fewer than 30 employees NT$1,000 (about US$31) for each day of parental leave taken one day at a time, roughly a day of minimum-wage earnings. A payment of this size covers the direct cost of an absence and may be enough to offset what the reform cost the smallest, lowest-paying firms. However, it does little to address the broader cost that firms appear to attach to hiring women of childbearing age. Our results suggest that a leave mandate can work against the workers it is meant to protect when employers expect childbirth to cost them more than the leave

itself. Narrowing the hiring gap would require policies that change what employers expect to happen after a birth.

The main limitation of our analysis is that the tax records report annual earnings but neither hours nor monthly employment histories, so we cannot separate low wages from part-time work, and we do not estimate the full equilibrium. For this reason, the cost estimates in Section 4 should be read as approximate, and those for the reform-induced response as upper bounds. Future research should estimate the full model with survey data on hours and quarterly employment transitions. Such estimates would show which costs firms associate with childbirth and how much a larger or longer subsidy would change their hiring.

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Appendices

A Background Details

Table 9: Taiwan's Parental Leave System: Provisions Before and After the 2009 Reform

	Pre-reform Mar 2002 – Apr 2009	**2009 reform** Effective May 1, 2009	**Subsequent amendments** (outside estimation window)
Panel A. Job-protected leave (Gender Equality in Employment Act, Art. 16)			
Firm-size eligibility	Firms with 30 or more employees	All firms, no size threshold	—
Tenure requirement	≥ 1 year with current employer	≥ 1 year	Reduced to ≥ 6 months (Dec 2014)
Child age limit	Child under age 3	Child under age 3	—
Maximum duration	Up to 2 years, ending when child turns 3	Same	Same
Reinstatement right	Yes, for covered employees	Yes, all employees	—
Spousal employment condition	Not available if spouse not employed (Art. 22)	Same	Art. 22 repealed (Jan 18, 2022)
Social insurance during leave	Coverage continues; employer premium waived, employee premium deferrable 3 years	Same	Same
Panel B. Leave allowance (Employment Insurance Act, Art. 11 & 19-2)			
Cash allowance	None (unpaid leave)	60% of average insured monthly wage over prior 6 months	Additional 20% wage subsidy, 80% total (Jul 1, 2021)
Employment insurance tenure requirement	—	≥ 1 year of cumulative Employment insurance coverage	—
Duration of allowance	—	6 months per child per parent	—
Combined parental limit	—	Up to 12 months per child; benefit periods may not overlap	Simultaneous receipt by both parents permitted (Jul 1, 2021)
Benefit cap	—	Insured wage capped at NT$43,900 (2009), implying a maximum of NT$26,340 per month	Cap raised periodically
Financing	—	Employment insurance fund (employee, employer, government contributions)	—

Notes: The removal of the 30-employee threshold was promulgated on January 16, 2008, but Article 40 delegated its effective date to the Executive Yuan, which set it at May 1, 2009 to coincide with the introduction of the employment-insurance allowance. Because leave may be taken until the child turns three, workers whose children were born before May 2009 could also become eligible, so the 2009 birth cohort is only partially exposed and is excluded from the cohort design in Section 4. Sources: Gender Equality in Employment Act, Employment Insurance Act, and Ministry of Labor implementing orders.

B Data Details

B.1 Data Construction and Administrative Record Linkage

For individuals with missing demographic information (such as gender or birth year) in specific annual records, we perform cross-record validation by retrieving these time-invariant characteristics from other available records associated with the same unique individual identifier. Within a household ID, when multiple individuals are recorded as spouses, we impute the individual closest in age as the spouse. To account for nonemployment, which may lead individuals not to file a tax return which would generate missing observations, we fill in any missing years between the first and last observed appearances for individuals who ever appear in the tax records with zeros. This approach implicitly assumes no employment, no labor force participation, and no wage income in those missing years. This approach captures the extensive margin of labor supply, under the assumption that a missing year reflects non-participation or non-employment and not data attrition.

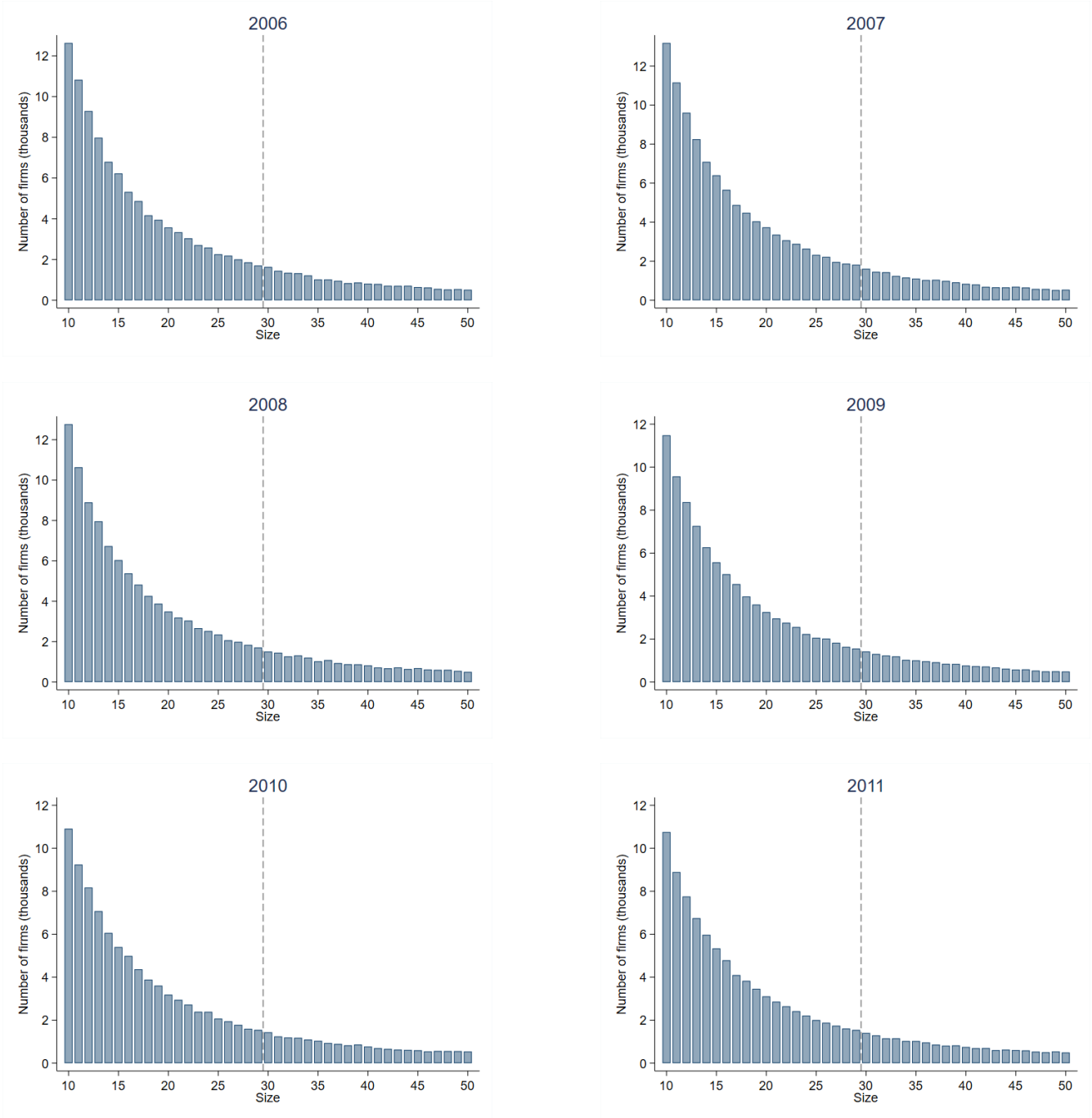


Figure 9: Firm size distribution

Note: Each panel in this figure plots the distribution of firm size around the 30-employee threshold (dashed gray line), by year from 2006 to 2017.

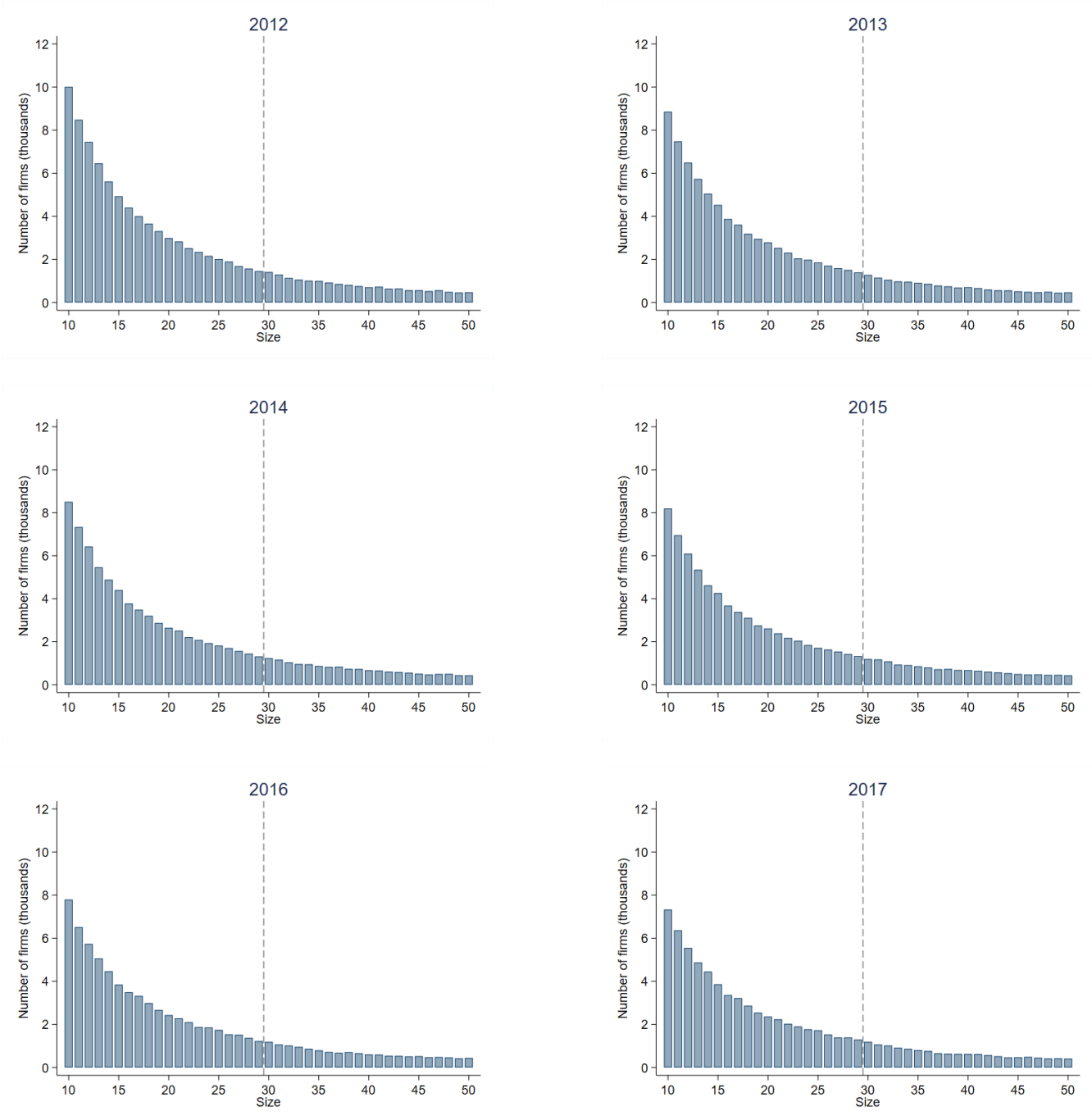
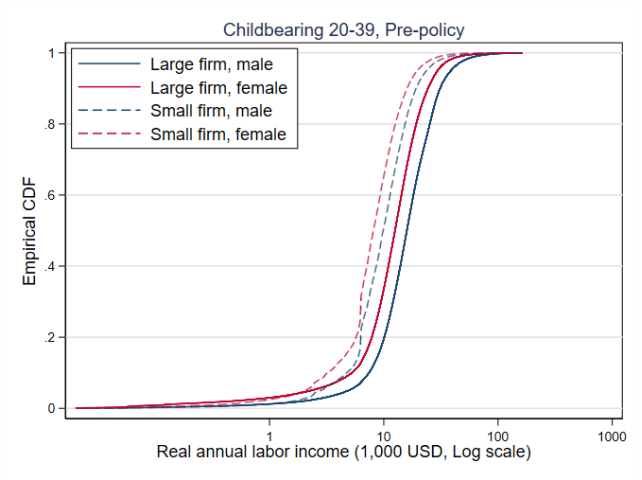
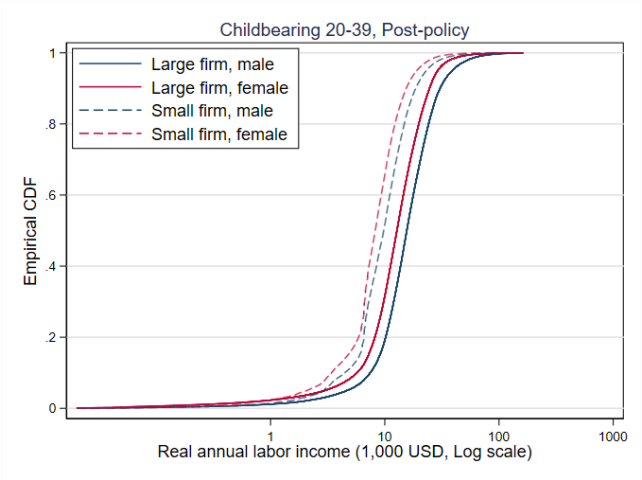


Figure 10: Firm size distribution (cont.)

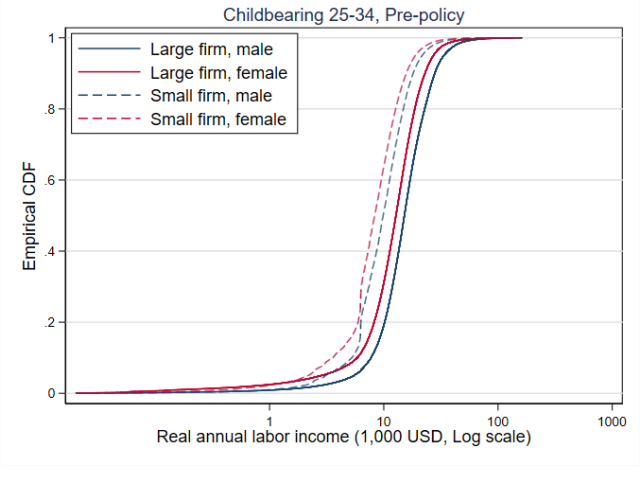
Note: Each panel in this figure plots the distribution of firm size around the 30-employee threshold (dashed gray line), by year from 2006 to 2017.



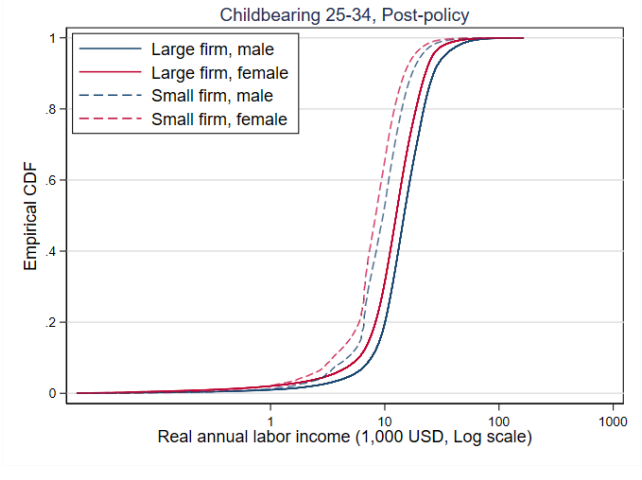
(a) Childbearing Age (20–39), pre-policy



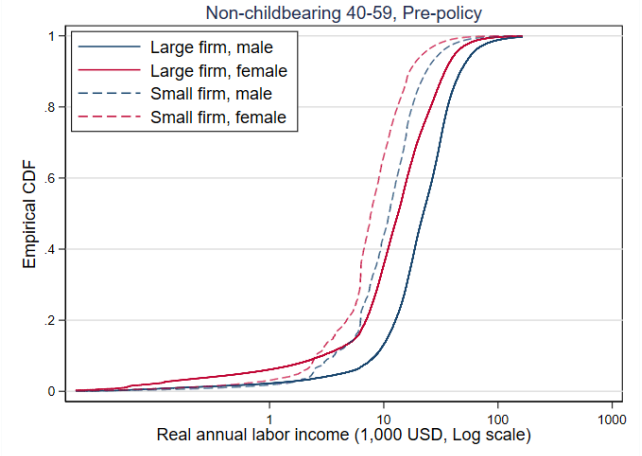
(b) Childbearing Age (20–39), post-policy



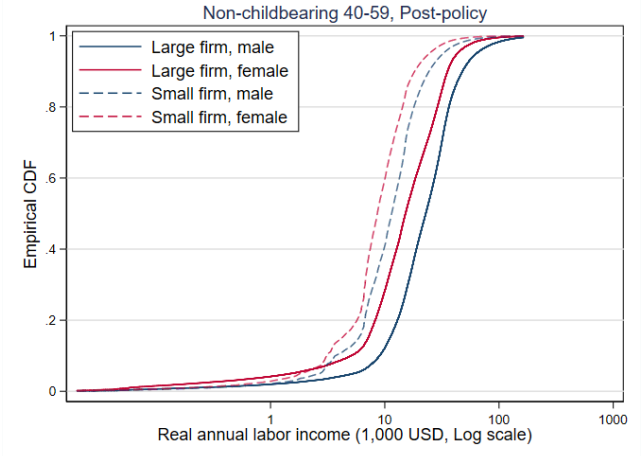
(c) Childbearing Age (25–34), pre-policy



(d) Childbearing Age (25–34), post-policy



(e) Non-childbearing Age (40–59), pre-policy



(f) Non-childbearing Age (40–59), post-policy

Figure 11: Empirical CDFs of real annual earnings by group, before and after the policy

Note: The x-axis is trimmed to the 1st–99th percentile for clarity, whereas the CDF uses the full sample.

Table 10: Kolmogorov-Smirnov tests for equality of pre- and post-policy earnings distributions

Age group	Subgroup	D	p-value
Childbearing age 20–39	Large firm, male	0.0255	< 0.001
	Large firm, female	0.0212	< 0.001
	Small firm, male	0.0534	< 0.001
	Small firm, female	0.0807	< 0.001
Non-childbearing age 40–59	Large firm, male	0.0160	< 0.001
	Large firm, female	0.0768	< 0.001
	Small firm, male	0.0559	< 0.001
	Small firm, female	0.1234	< 0.001
Childbearing age 25–34	Large firm, male	0.0253	< 0.001
	Large firm, female	0.0151	< 0.001
	Small firm, male	0.0478	< 0.001
	Small firm, female	0.0683	< 0.001

Notes: Each row reports the Kolmogorov-Smirnov D statistic and p-value for the two-sample test comparing the empirical distribution of real annual earnings before versus after the reform, separately by gender, firm size, and age group. Group labels follow the same ordering as the legend in the corresponding empirical CDF figures. All p-values are below 0.001 due to the large sample size; the magnitude of D indicates the maximum divergence between the pre- and post-policy CDFs is modest across all subgroups.

C Additional Figures

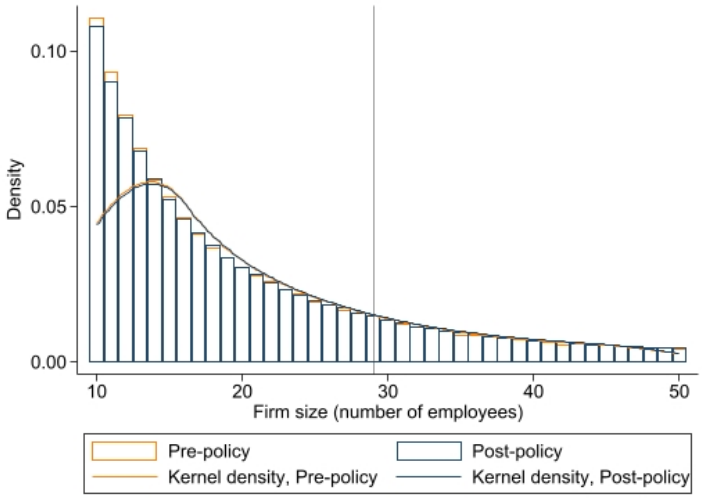
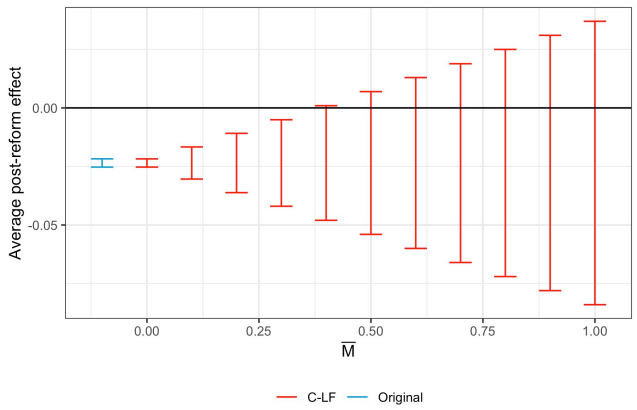
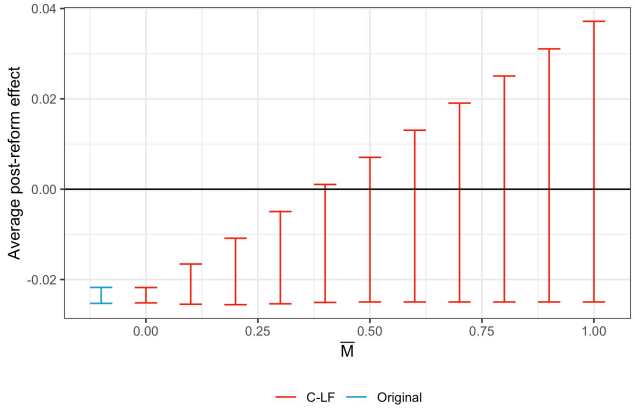


Figure 12: Firm size distribution, pre-policy and post-policy

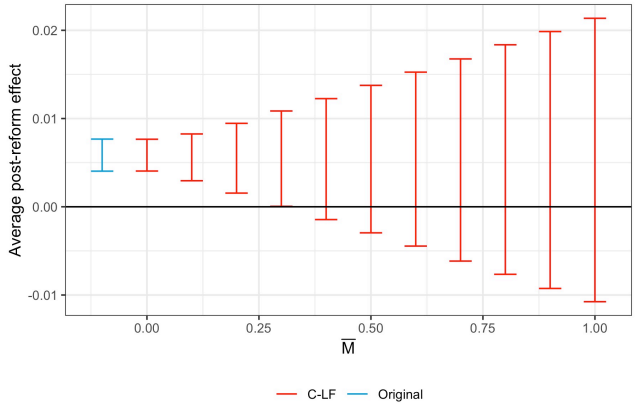
Note: This figure plots the firm size distribution for firms with 10 to 50 employees, separately for the pre-policy period (2006-2008) and the post-policy period (2010-2017). Bars show the share of firms at each firm size, with pre-policy in orange outline and post-policy in navy. Solid lines are kernel density estimates for each period. The vertical gray line marks the 30-employee threshold.



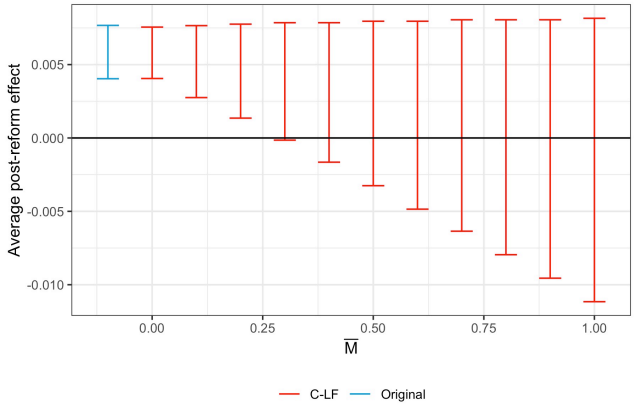
(a) J2J to small firms, no restriction



(b) J2J to small firms, negative bias



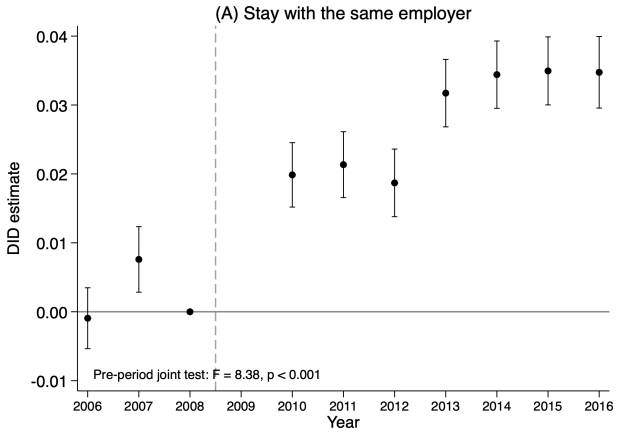
(c) J2J to large firms, no restriction



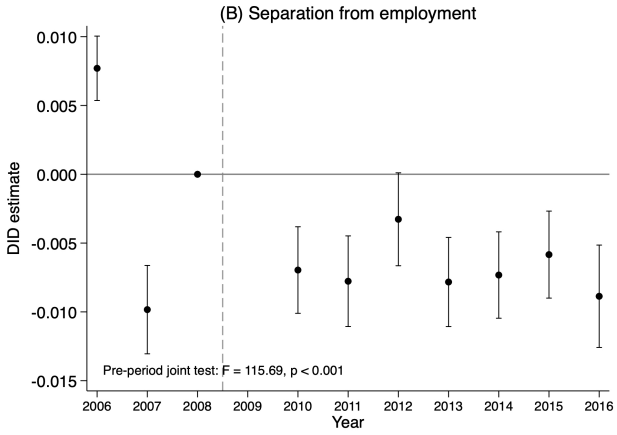
(d) J2J to large firms, positive bias

Figure 13: Sensitivity to Violations of Parallel Trends: Job-to-Job Transitions

Note: Each panel reports 95% robust confidence sets for the average post-reform effect (2010 to 2016) using the relative magnitudes approach of Rambachan and Roth (2023). $\bar{M}$ bounds the post-reform deviation from parallel trends relative to the largest consecutive change in the pre-reform coefficients. The blue interval is the original confidence interval, and the red intervals are robust confidence sets computed with the conditional-least favorable hybrid (C-LF) method. Sign restrictions impose that any bias is in the direction that would overstate the estimated effect. The reference year is 2008; 2009 is excluded.



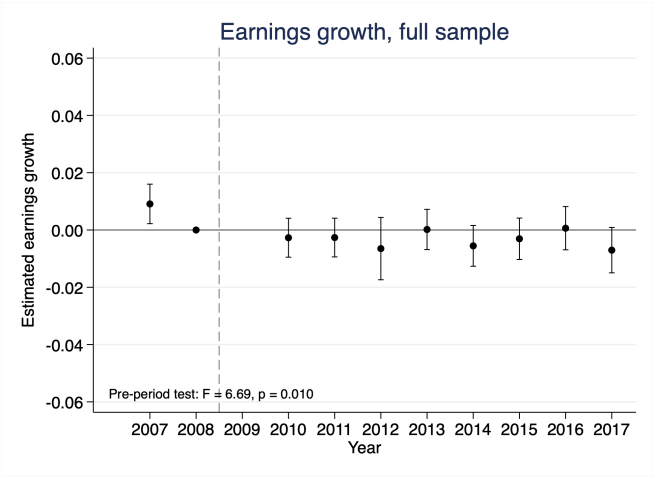
(a) Stay with the same employer



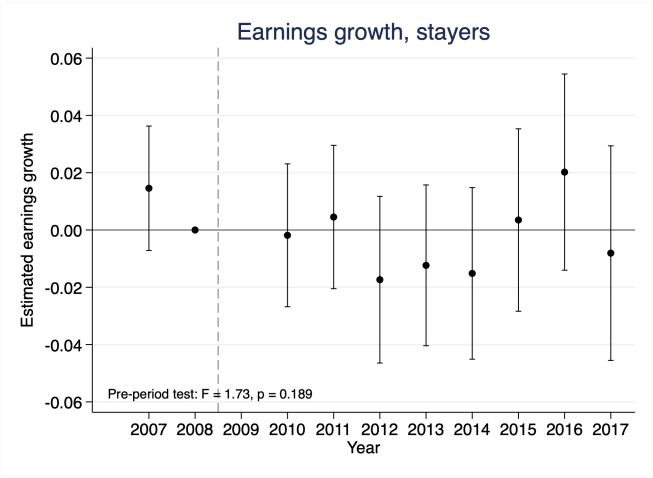
(b) Separation from employment

Figure 14: Event-Study Estimates: Stay and Separation

Note: This figure plots event-study estimates of the effect of the reform on the probability that childbearing-age women at small firms stay with the same employer (Panel A) and separate from employment (Panel B), relative to women at large firms. The reference year is 2008; 2009 is excluded. Bars show 95% confidence intervals. The pre-period joint test reports the F-statistic and p-value for the null that the 2006 and 2007 coefficients are jointly zero.



(a) Full Sample



(b) Job Stayers

Figure 15: Differential Earnings Growth: Full Sample vs. Job Stayers

Note: This figure plots the estimated β_k from equation (5), the change in the gap in log earnings growth between women and men at small firms relative to the same gap at large firms, for workers aged 20–39. Panel (a) includes all workers with positive earnings in both $t-1$ and t. Panel (b) restricts the sample to workers who stay with the same employer between $t-1$ and t. The reference year is 2008; 2009 is excluded. Standard errors are clustered at the firm level, and the figure shows 95 percent confidence intervals.

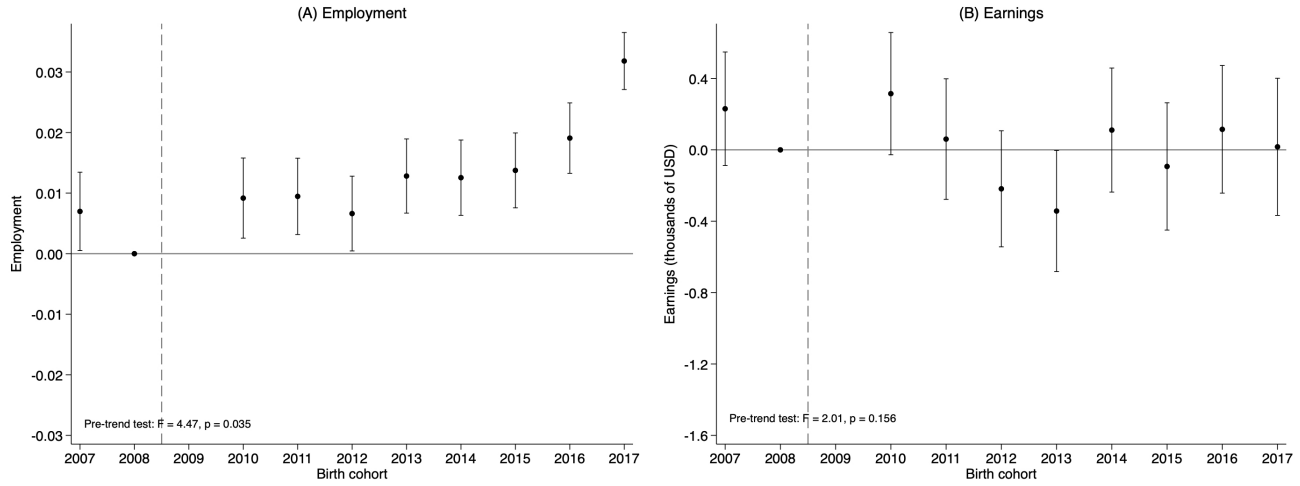


Figure 16: Pre-Reform Trends by Birth Cohort, Fathers

Notes: The figure plots estimates of $\beta_{c,s}$ from equation (7) at horizon $s = 0$ with 95% confidence intervals. Each coefficient is the employment (Panel A) or earnings (Panel B) gap between fathers at small and large firms in birth cohort c, relative to the same gap in the 2008 cohort. Treatment is defined by the father's employer in the year before birth. The dashed line marks the 2009 reform. The pre-trend test reports the F-statistic and p-value for $H_0 : \beta_{2007,s} = 0$. All regressions include calendar-year and age fixed effects. Standard errors are clustered at the level of the pre-birth employer.

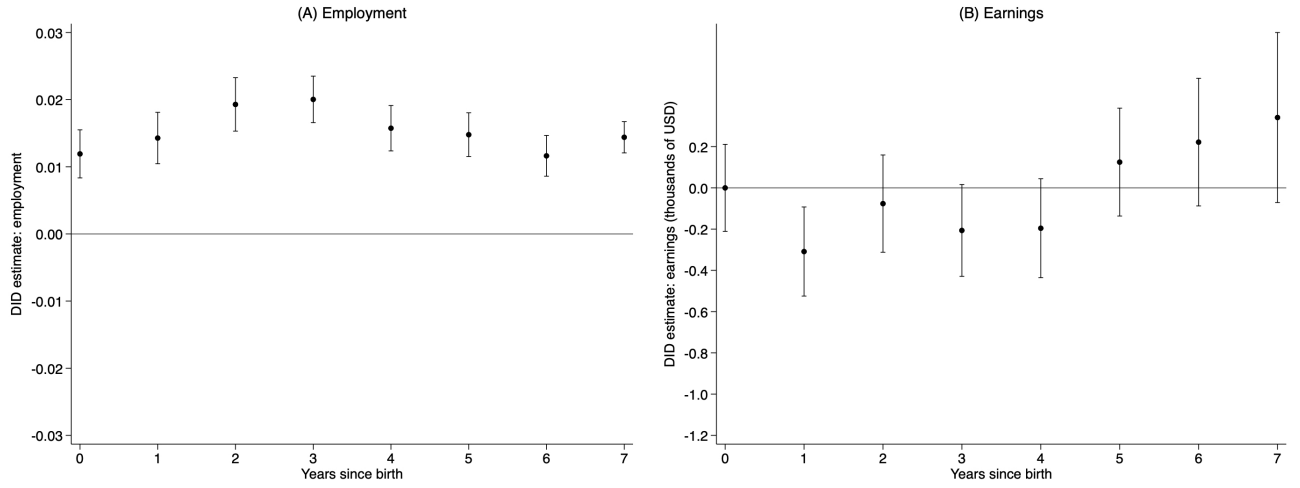


Figure 17: Difference-in-Differences Estimates by Years Since Birth, Fathers

Notes: Each point is $\hat{\theta}_s$ from a separate estimation of equation (6) at horizon s, with 95% confidence intervals. Panel A is employment; Panel B uses annual earnings in thousands of USD. All regressions include calendar-year and age fixed effects. Standard errors are clustered at the level of the pre-birth employer.

D Model Derivations and Proofs

The main text uses three restrictions of the model, namely the hiring rule in Proposition 1, the job-finding ratio in (11), and the inversion of the type-n wage distribution in (12). This appendix records the rest of the equilibrium, including the offer distributions of childbearing-age women, firm size, the worker's problem and reservation wages, steady-state stocks, the wage schedule, and the implied earnings distributions. None of these objects is needed for the calculations in Table 8. We report them to show that the restrictions used in the main text come from a complete equilibrium, and because a full estimation of the model would build on them. Sections D.8–D.11 describe the estimation of k, the treatment of the minimum wage, the evidence on firm size, and the sensitivity of the results.

D.1 Equilibrium

Let $m(p) \equiv \ell_n\big(w(p)\big) - 1$ denote the number of type-n coworkers at a firm with productivity p, and let $h(p) \equiv \mathbb{1}\{(1+m(p))^{\eta}\rho(p) \geq \bar{\Delta}\}$ indicate whether the firm hires type-y workers. Define the share of firms that do not hire them, $H \equiv \int [1-h(p)]\, d\Gamma(p)$, and the share of hiring firms with productivity above p, $S(p) \equiv \int_p^{\bar{p}} h(x)\, d\Gamma(x)$.

Definition 1 (Steady-state equilibrium)**.** Given primitives $\{r, \lambda_0, \lambda_1, \delta_n, \delta_y, \kappa, \gamma, \nu, b_n, b_y, \varsigma, \bar{c}, \eta, \Gamma, N_n, N_y\}$, a steady-state equilibrium consists of reservation wages (R_n, R_y), a wage schedule $w(\cdot)$ on $[\underline{p}, \bar{p}]$, a hiring rule $h(\cdot)$, offer distributions (F, F_y) with effective arrival rates $(\tilde{\lambda}_{0y}, \tilde{\lambda}_{1y})$, firm-level employment (ℓ_n, ℓ_y), and stocks (u_n, u_y, e_y, n_y) such that

1. *Workers optimize.* Given (F, F_y) and $(\tilde{\lambda}_{0y}, \tilde{\lambda}_{1y})$, unemployed workers of type g accept any offer at or above R_g, employed workers accept any offer above their current wage, and (R_n, R_y) satisfy Proposition 2.

2. *Firms optimize.* Each firm posts $w(p)$ to maximize $(p-w)\ell_n(w)$ and pays the same wage to every worker it hires. It hires type-y workers if and only if $h(p) = 1$ (Proposition 1).

3. *Offers are consistent with wage posting.* $F(w(p)) = \Gamma(p)$, and F_y, $\tilde{\lambda}_{0y}$, and $\tilde{\lambda}_{1y}$ are given by (10) and (14).

4. *Flows balance.* Firm-level employment satisfies (16)–(17), and the stocks satisfy Proposition 3.

When the breakeven cost $z(p) = (1+m(p))^{\eta}\rho(p)$ is increasing in p, the hiring rule is a threshold, $h(p) = \mathbb{1}\{p \geq p^{**}\}$, so that $H = \Gamma(p^{**})$ and $S(p) = 1 - \Gamma(p)$ for $p \geq p^{**}$. The expressions below hold in general and simplify accordingly in this case.

D.2 Offer distributions

Derivation of (10). Contacts with firms arrive to a type-y worker as a Poisson process of rate λ_0 when she is unemployed and λ_1 when she is employed and at work, with each contacted firm's productivity an independent draw from Γ. A contact becomes an offer only if the firm hires type-y workers, $h(p) = 1$, which occurs with probability $1 - H$ independently of the arrival process. By the thinning property of Poisson processes, if each event of a Poisson process with rate λ is retained independently with probability q, the retained events form a Poisson process with rate λq. Applying this to the unemployed and employed contact processes gives $\tilde{\lambda}_{jy} = \lambda_j(1 - H)$ for $j \in \{0, 1\}$. Every offer pays the posted wage $w(p)$, and since $w(\cdot)$ is strictly increasing (Proposition 4), the distribution of the wages offered to type-y women is

$$F_y\big(w(p)\big) = \frac{1}{1 - H} \int_{\underline{p}}^{p} h(x)\, d\Gamma(x). \tag{14}$$

□

An identity used repeatedly below follows directly. For a type-y woman employed at a firm with productivity p,

$$\tilde{\lambda}_{1y}\, \bar{F}_y\big(w(p)\big) = \lambda_1 S(p). \tag{15}$$

She is poached at the rate at which she meets hiring firms that pay more than her current employer. Under a threshold rule, $S(p) = 1 - \Gamma(p)$ for every $p \geq p^{**}$, so her poaching rate equals that of a type-n worker at the same firm.

D.3 Firm size

Given its posted wage, a firm's steady-state stock of each type takes the Burdett–Mortensen form

$$\ell_n(p) = \frac{D_n}{A_n(p)^2}, \qquad \ell_y(p) = h(p)\, \frac{D_y}{A_y(p)^2}, \qquad A_n(p) \equiv \delta_n + \lambda_1\big[1 - \Gamma(p)\big], \quad A_y(p) \equiv \delta_y + \lambda_1 S(p), \tag{16}$$

where $A_g(p)$ is the rate at which the firm loses a type-g worker who is at work, through separation into non-employment and through poaching. The constants are

$$D_n = N_n\, u_n\, \lambda_0\big(\delta_n + \lambda_1\big), \qquad D_y = \frac{N_y\, u_y\, \lambda_0\big(\delta_y + \tilde{\lambda}_{1y}\big)}{1 - \phi}, \tag{17}$$

with u_g given in Proposition 3.

Leave and the firm's exit rate. A woman on leave is attached to her job but exposed to neither exit channel. Job protection shields her from separation, and she does not search, so she cannot be poached. Among the type-y workers attached to firms posting a given wage, a worker at work goes on leave at rate κ and a worker on leave returns at rate γ, so a share $\phi = \kappa/(\kappa+\gamma)$ of attached workers is on leave at every wage. As a result, the attached pool leaves the firm at rate $(1-\phi)A_y(p)$, and $1-\phi$ enters D_y without changing the shape of ℓ_y.

Derivation of (16)–(17). The construction follows Burdett and Mortensen (1998, Section 2). We write the steady-state flow balance for the stock of workers earning w or less, then take its derivative with respect to the mass of firms.

Type n. Let $L_n(w)$ denote the mass of type-n workers employed at firms posting wages at or below w. Since $F(R_n) = 0$, the flow balance is

$$\lambda_0 \, F(w) \, u_n N_n = \big[\delta_n + \lambda_1 \bar{F}(w)\big] L_n(w), \tag{18}$$

so $L_n(w) = \lambda_0 u_n N_n F(w)/[\delta_n + \lambda_1 \bar{F}(w)]$. Because F is the distribution of offers across the whole continuum of firms, employment per firm is $\ell_n = dL_n/dF$, which gives the expression in (16) after using $F(w(p)) = \Gamma(p)$.

Type y. Let $L_y(w)$ denote the mass of attached type-y workers, at work or on leave, at hiring firms posting wages at or below w. The flow balance parallels (18) with three changes. The arrival rate is $\tilde{\lambda}_{0y}$, the offer distribution is F_y, and the outflow applies only to the share $1-\phi$ of attached workers who are at work,

$$\tilde{\lambda}_{0y} \, F_y(w) \, u_y N_y = (1-\phi)\big[\delta_y + \tilde{\lambda}_{1y} \bar{F}_y(w)\big] L_y(w).$$

A unit of F_y corresponds to $1-H$ units of firm mass, so employment per hiring firm is

$$\ell_y = \frac{1}{1-H}\frac{dL_y}{dF_y} = \frac{1}{1-H}\,\frac{\tilde{\lambda}_{0y}\, u_y N_y (\delta_y + \tilde{\lambda}_{1y})}{(1-\phi)\big[\delta_y + \tilde{\lambda}_{1y}\bar{F}_y(w)\big]^2}.$$

Substituting $\tilde{\lambda}_{0y} = \lambda_0(1-H)$ cancels the leading factor, and (15) gives the denominator in terms of $S(p)$. □

D.4 Worker's problem

Let V_0^g denote the value of unemployment, $V_1^g(w)$ the value of working at wage w, and $M^y(w)$ the value of being on leave from a job paying w. For a type-n worker,

$$rV_0^n = b_n + \lambda_0 \int \max\{0, V_1^n(x) - V_0^n\}\, dF(x), \tag{19}$$

$$rV_1^n(w) = w - \varsigma + \lambda_1 \int_w^{\bar{w}} \big[V_1^n(x) - V_1^n(w)\big] dF(x) + \delta_n\big(V_0^n - V_1^n(w)\big). \tag{20}$$

For a type-y woman,

$$rV_0^y = b_y + \tilde{\lambda}_{0y} \int \max\{0, V_1^y(x) - V_0^y\}\, dF_y(x), \tag{21}$$

$$rV_1^y(w) = w - \varsigma + \tilde{\lambda}_{1y} \int_w^{\bar{w}} \big[V_1^y(x) - V_1^y(w)\big] dF_y(x) + \kappa\big(M^y(w) - V_1^y(w)\big) + \delta_y\big(V_0^y - V_1^y(w)\big), \tag{22}$$

$$rM^y(w) = \nu + \gamma\big(V_1^y(w) - M^y(w)\big), \tag{23}$$

so that $M^y(w) = \big(\nu + \gamma V_1^y(w)\big)/(r+\gamma)$.

Lemma 1 (Marginal value of a job)**.** *V_1^g is strictly increasing in w, with*

$$V_1^{n\prime}(w) = \frac{1}{r + \delta_n + \lambda_1 \bar{F}(w)}, \qquad V_1^{y\prime}(w) = \frac{1}{r + \delta_y + \tilde{\lambda}_{1y}\bar{F}_y(w) + \kappa_{PL}}, \qquad \kappa_{PL} \equiv \frac{\kappa r}{r+\gamma}.$$

Proof. Differentiate (22) with respect to w, using $M^{y\prime}(w) = \gamma V_1^{y\prime}(w)/(r+\gamma)$ from (23),

$$rV_1^{y\prime} = 1 - \tilde{\lambda}_{1y}\bar{F}_y(w) V_1^{y\prime} - \kappa\Big(1 - \frac{\gamma}{r+\gamma}\Big) V_1^{y\prime} - \delta_y V_1^{y\prime},$$

and collect terms. The type-n case is the same with $\kappa = 0$. □

Proposition 2 (Reservation wages)**.** *Let $I_n \equiv \int_{R_n}^{\bar{w}} \frac{\bar{F}(x)}{r+\delta_n+\lambda_1\bar{F}(x)}\, dx$ and $I_y \equiv \int_{R_y}^{\bar{w}} \frac{\bar{F}_y(x)}{r+\delta_y+\tilde{\lambda}_{1y}\bar{F}_y(x)+\kappa_{PL}}\, dx$. Then*

$$R_n - b_n - \varsigma = (\lambda_0 - \lambda_1)\, I_n,$$

$$R_y - b_y - \varsigma = (\tilde{\lambda}_{0y} - \tilde{\lambda}_{1y})\, I_y + \frac{\kappa}{r+\gamma}\Big(b_y + \tilde{\lambda}_{0y} I_y - \nu\Big).$$

Proof. At $w = R_g$ the worker is indifferent, $V_1^g(R_g) = V_0^g$. Integrating by parts with Lemma 1, $\int_{R_y}^{\bar{w}} [V_1^y(x) - V_0^y]\, dF_y(x) = I_y$, so (21) gives $rV_0^y = b_y + \tilde{\lambda}_{0y} I_y$. Evaluating (22) at R_y, where $M^y(R_y) - V_1^y(R_y) = (\nu - rV_0^y)/(r+\gamma)$, gives

$$rV_0^y = R_y - \varsigma + \tilde{\lambda}_{1y} I_y + \frac{\kappa}{r+\gamma}\big(\nu - rV_0^y\big).$$

Equating the two expressions for rV_0^y yields the result. The type-n case is the same with $\kappa = 0$. □

When the leave benefit ν falls short of the flow value of unemployment, rV_0^y, taking a job carries the prospect of a spell on leave that is worth less than continued search, which raises R_y. Because the hiring rule depends only on firm rents and the number of coworkers, R_y does not enter it. We assume that R_y lies below the lowest wage paid by a hiring firm, so every offer is accepted, as used in (11). This holds whenever the reservation wage is below the minimum wage.

D.5 Steady state

Proposition 3 (Steady-state distribution)**.** *For type n, $u_n = \delta_n/(\delta_n + \lambda_0)$. For type-$y$ women, the shares unemployed, at work, and on leave are*

$$u_y = \frac{\delta_y(1-\phi)}{\tilde{\lambda}_{0y} + \delta_y(1-\phi)}, \qquad e_y = \frac{\tilde{\lambda}_{0y}(1-\phi)}{\tilde{\lambda}_{0y} + \delta_y(1-\phi)}, \qquad n_y = \frac{\tilde{\lambda}_{0y}\,\phi}{\tilde{\lambda}_{0y} + \delta_y(1-\phi)}.$$

Proof. Flows out of unemployment, $\tilde{\lambda}_{0y}u_y$, equal separations of women at work, $\delta_y e_y$. Flows into leave, κe_y, equal returns, γn_y, so $n_y = e_y\,\phi/(1-\phi)$. Combining with $u_y + e_y + n_y = 1$ gives the result. □

Attached employment, $e_y + n_y = \tilde{\lambda}_{0y}/[\tilde{\lambda}_{0y} + \delta_y(1-\phi)]$, depends on the hiring rule through $\tilde{\lambda}_{0y}$. The policy affects employment as well as composition, because leave shields attached women from separation while the hiring rule lowers their offer arrival rate.

D.6 The wage schedule

Proposition 4 (Posted wage)**.** *The wage schedule $w(\cdot)$ solves the single-type Burdett–Mortensen equation*

$$w'(p) = \frac{2\lambda_1\Gamma'(p)\,\big(p - w(p)\big)}{\delta_n + \lambda_1[1-\Gamma(p)]},$$

with boundary condition $w(\underline{p}) = R_n$. Neither $\bar{c}$, η, nor ϕ enters, so $w(\cdot)$ is invariant to the leave policy.

Proof. The firm chooses w to maximize $(p-w)\ell_n(w)$. By (16), the first-order condition is $(p-w)\,2\lambda_1 f(w)/A_n = 1$. Along the equilibrium schedule, $F(w(p)) = \Gamma(p)$, so $f(w(p))\,w'(p) = \Gamma'(p)$. Substituting gives the result. As in Burdett and Mortensen (1998), the lowest-productivity firm pays the reservation wage. Combining the first-order condition with $G = F/(1+k\bar{F})$ gives the rent in (12) (Bontemps et al., 2000). □

Under equal pay, a firm that hires childbearing-age women pays them $w(p)$. Their margin is then $(1-\phi)\rho(p) - \phi c(m)$, which gives the hiring rule in (9).

D.7 Earnings distributions and predictions

Proposition 5 (Earnings distributions)**.** *The wage distributions of type-g workers at work are*

$$G_n(w) = \frac{F(w)}{1 + k_n \bar{F}(w)}, \qquad G_y(w) = \frac{F_y(w)}{1 + k_y \bar{F}_y(w)}, \qquad k_n = \frac{\lambda_1}{\delta_n}, \qquad k_y = \frac{\tilde{\lambda}_{1y}}{\delta_y}.$$

Proof. $G_g(w) = L_g(w)/L_g(\bar{w})$. For type n, (18) gives $L_n(\bar{w}) = \lambda_0 u_n N_n/\delta_n$, and dividing yields the result. For type y, the factor $1-\phi$ appears in both $L_y(w)$ and $L_y(\bar{w})$ and cancels, and the share on leave is the same at every wage. □

The model delivers three predictions that we take to the data or use to interpret it.

1. *Hiring among small, low-wage firms.* Firms with low rents and few coworkers stop hiring childbearing-age women. Extending the cost of leave to small firms lowers their hiring of these women, and within small firms the decline should be concentrated among those with the fewest employees and the lowest wages. Appendix Table 11 shows that the female-specific decline is larger at firms with fewer employees.

2. *No wage response.* Because $w(\cdot)$ is set by the type-n margin and equal pay applies to all workers in a firm, the leave cost does not change any worker's wage, including the wages of childbearing-age women already employed. Firms adjust only through hiring.

3. *Job-to-job moves.* Under a threshold rule, the identity in (15) implies that a type-y woman at a given employer is poached at the same rate as a type-n worker at that employer. Under a threshold rule, the model does not predict a change in job-to-job moves conditional on the employer. Fewer moves into small firms can arise when firms that pay similar wages differ in size, so that some small firms paying more than a woman's current employer stop hiring her, but this lies outside the baseline model.

D.8 Estimating k

Annual transition rates. In the Burdett–Mortensen steady state, the average job-to-job rate of employed type-n workers is

$$\bar{\jmath}_n = \lambda_1 \int \bar{F}(w)\, dG_n(w) = \delta_n \left[\frac{(1+k)\ln(1+k)}{k} - 1 \right],$$

so k can in principle be recovered from the ratio $\bar{\jmath}_n/\delta_n$. We estimate the monthly generator of the three-state process (same employer, different employer, non-employment) from the annual transition matrix as

$\hat{Q} = \frac{1}{12}\log \hat{P}_{\text{annual}}$. The implied monthly rates for type-n workers are $\hat{\delta}_n = 0.0030$ and $\hat{j}_n = 0.0219$, a ratio of 7.2, and inverting the expression above gives $\hat{k} \approx 3{,}700$. These values are far outside the range estimated in the literature, and the reason is mechanical. The right-hand side rises only logarithmically in k. A ratio of 0.39 corresponds to $k = 1$, a ratio of 1.64 to $k = 10$, and a ratio of 5.9 to $k = 1{,}000$. As a result, small biases in the ratio produce very large biases in k, and two biases push the ratio up. Annual data miss non-employment spells that begin and end within a calendar year, so a worker who loses her job and finds another within the year is recorded as a job-to-job mover. In addition, many job-to-job moves are driven by layoffs, relocation, or family reasons and need not move workers up the wage ladder. Jolivet et al. (2006) model these moves as reallocation shocks.

Wage distributions. We instead estimate k from (13), which relates the wage distribution of full-year stayers, G, to the offer distribution, F. We measure G from 7,354,315 worker-years of type-n workers employed by the same eligible employer in $t-1$, t, and $t+1$ in 2010–2016, with annual earnings at or above that year's minimum wage and below the 99th percentile. We measure F from type-n workers hired out of non-employment in 2010–2015 who stay with the new employer through $t+2$, using their earnings in $t+1$, the first full calendar year. Hires that return to a recent employer are excluded. We compute both distributions in NT\$5,000 bins and estimate k by least squares without a constant over bins above w_b with $G < 0.99$. A worker is hired out of non-employment if she had no employer in $t-1$, including workers who first appear in the data in t, which yields 1,795,362 hires. The estimate is $\hat{k} = 0.44$ (s.e. 0.010), at the lower end of the range in Jolivet et al. (2006). The estimate is not sensitive to the choice of w_b between 1.2 and 1.4 times the minimum wage.

D.9 Rents Near the Minimum Wage

Annual earnings of full-year type-n workers bunch at the minimum wage. Expressed relative to the 2015 annual minimum wage, the spike spans roughly 0.95 to 1.03, because the minimum wage varied across years in real terms, and the density levels off by about 1.2. We set $w_b = 1.2$ times the annual minimum wage, below which 11 percent of full-year type-n workers fall. At $w_b = 1.4$, this share rises to 21.5 percent.

Above w_b, the density g in (12) must be estimated on a sample truncated at w_b. A kernel estimator sees data only to the right of the truncation point and underestimates the density there. In our data, the kernel estimate rises from 1.28×10^{-6} at w_b + NT\$5,000 to 2.14×10^{-6} at w_b + NT\$25,000, and the implied rent at w_b is NT\$934,894. The histogram estimator we use, based on counts in NT\$5,000 bins smoothed over three bins, is not affected by the truncation and gives a rent at w_b of NT\$521,181. Away

from the truncation point, the two estimators agree.

D.10 Hiring Responses by Detailed Firm Size

Table 11: Hiring responses by detailed size of small firms

	Women 20–39	Women − Men, 20–39
	(1)	(2)
3–5 employees × post	-0.0087***	-0.0082***
	(0.0016)	(0.0027)
6–9 employees × post	-0.0112***	-0.0077***
	(0.0016)	(0.0026)
10–14 employees × post	-0.0140***	-0.0102***
	(0.0016)	(0.0028)
15–19 employees × post	-0.0044***	-0.0067***
	(0.0014)	(0.0023)
20–24 employees × post	-0.0142***	-0.0063*
	(0.0021)	(0.0036)
25–29 employees × post	-0.0148***	-0.0031
	(0.0024)	(0.0040)
p-value: 3–5 = 25–29		0.244
p-value: 3–14 = 15–29		0.001
Size group FE	Y	Y
Industry × year FE	Y	Y
N	2,530,010	2,530,010

Notes: The outcome is the share of women aged 20–39 among new hires (column 1) or that share minus the share of men aged 20–39, computed for each firm and year (column 2). Each column reports one regression in which small firms are split by their 2007 employment, with large firms (30 to 264 employees in 2007) as the omitted group. The second p-value tests whether the sum of the three coefficients for firms with 3–14 employees equals the sum for firms with 15–29 employees. Standard errors are clustered at the firm level and reported in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

D.11 Sensitivity of the Implied Leave Cost

Table 12 reports the mean wedge for alternative values of k and of the lower cutoff w_b. A larger k lowers the rent in (12) and with it the implied wedge, while a higher w_b assigns more firms the rent at w_b. The implied wedge is roughly proportional to $1/k$, and moving from $k = 0.44$ to $k = 2$ lowers it by

about two thirds in every column. The cutoff w_b matters much less. Raising it from 1.2 to 1.4 times the minimum wage changes the wedge for the raw gap by less than 10 percent and lowers the wedge for the reform-induced gap by about 10 to 15 percent. Across all specifications, the wedge for the raw gap stays at or above 0.8 times the annual minimum wage, while the mean wedge for the reform-induced gap falls to about a quarter of the minimum wage when $k = 2$ and $\eta = 1$.

Table 12: Sensitivity of the implied leave cost

		Raw gap, all firms			Reform-induced, small firms		
w_b	k	$\eta = 0$	$\eta = 0.5$	$\eta = 1$	$\eta = 0$	$\eta = 0.5$	$\eta = 1$
1.2	0.44	2.70	2.76	3.20	2.18	1.46	1.03
1.2	1	1.37	1.39	1.62	1.07	0.69	0.49
1.2	2	0.84	0.85	1.00	0.60	0.40	0.28
1.4	0.44	2.73	2.57	2.99	1.96	1.26	0.89
1.4	1	1.38	1.32	1.52	0.97	0.62	0.44
1.4	2	0.84	0.81	0.96	0.57	0.37	0.26
Direct cost of absence, Δ^{direct} / min. wage							
NT$1,000 per leave day: 0.014					NT$2,000 per leave day: 0.028		

Notes: Each entry is the mean implied wedge $\hat{\Delta}(m)$ relative to the annual minimum wage of NT$230,505 (US$7,226 at the 2015 exchange rate of 31.898), across all firms (raw gap, $\hat{R} = 0.579$) or small firms (reform-induced, $\hat{R}^{\text{DiD}} = 0.964$). w_b is the lower cutoff of the wage distribution, in multiples of the annual minimum wage. The first row is the baseline in Table 8. With $w_b = 1.4$, 21.5 percent of full-year type-n workers and 34.1 percent of small firms fall below the cutoff, so the reform-induced quantile lies inside the minimum-wage region and equals the rent at w_b (scaled by $(1 + m)^{\eta}$).